EVB-LAN9696-24port Hardware Manual

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Preface

NOTICE TO CUSTOMERS

All documentation becomes dated, and this manual is no exception. Microchip tools and documentation are constantly evolving to meet customer needs, so some actual dialogs and/or tool descriptions may differ from those in this document. Please refer to our web site (www.microchip.com) to obtain the latest documentation available.

Documents are identified with a "DS" number. This number is located on the bottom of each page, in front of the page number. The numbering convention for the DS number is "DSXXXXXA", where "XXXXX" is the document number and "A" is the revision level of the document.

For the most up-to-date information on development tools, see the MPLAB® IDE online help. Select the Help menu, and then Topics to open a list of available online help files.

INTRODUCTION

This chapter contains general information that will be useful to know before using the EVB-LAN9696-24port Hardware Manual. Items discussed in this chapter include:

- Document Layout
- Conventions Used in this Guide
- Warranty Registration
- The Microchip Web Site
- Development Systems Customer Change Notification Service
- Customer Support
- Document Revision History

DOCUMENT LAYOUT

The manual layout is as follows:

- **Chapter 1. "Overview"** – This chapter provides an overview of the additional documentation needed.
- **Chapter 2. "EVB-LAN9696 Reference Board"** – This chapter provides a brief overview of the EVB-LAN9696 and its main features.
- **Chapter 3. "Management Software"** – This chapter provides information on the software management of the EVB-LAN9696 Reference Board.
- **Chapter 4. "LED Indicators and Board Connector"** – This chapter is a short walk-through on identifying board connectors and LEDs.
- **Chapter 5. "Detailed Hardware Description"** – This section describes the hardware details of the EVB-LAN9696 Reference Board.
- **Chapter 6. "Environmental Requirements"** – This section describes the environmental requirements.
- **Appendix A. "PCB Layout"** – This section shows the different PCB layers, espe-

cially the routing to the DDR RAM.

CONVENTIONS USED IN THIS GUIDE

This manual uses the following documentation conventions:

DOCUMENTATION CONVENTIONS

Description	Represents	Examples
Arial font:		
Italic characters	Referenced books	*MPLAB® IDE User's Guide*
	Emphasized text	...is the *only* compiler...
Initial caps	A window	the Output window
	A dialog	the Settings dialog
	A menu selection	select Enable Programmer
Quotes	A field name in a window or dialog	"Save project before build"
Underlined, italic text with right angle bracket	A menu path	<u>*File>Save*</u>
Bold characters	A dialog button	Click **OK**
	A tab	Click the **Power** tab
N'Rnnnn	A number in verilog format, where N is the total number of digits, R is the radix and n is a digit.	4'b0010, 2'hF1
Text in angle brackets < >	A key on the keyboard	Press <Enter>, <F1>
Courier New font:		
Plain Courier New	Sample source code	`#define START`
	Filenames	`autoexec.bat`
	File paths	`c:\mcc18\h`
	Keywords	`_asm, _endasm, static`
	Command-line options	`-Opa+, -Opa-`
	Bit values	`0, 1`
	Constants	`0xFF, 'A'`
Italic Courier New	A variable argument	*`file`*`.o`, where *`file`* can be any valid filename
Square brackets [ ]	Optional arguments	`mcc18 [options]` *`file`* `[options]`
Curly brackets and pipe character: { | }	Choice of mutually exclusive arguments; an OR selection	`errorlevel {0|1}`
Ellipses...	Replaces repeated text	`var_name [, var_name...]`

	Represents code supplied by user	`void main (void)`
		`{ ...`
		`}`

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- **Product Support** – Data sheets and errata, application notes and sample programs, design resources, user's guides and hardware support documents, latest software releases and archived software
- **General Technical Support** – Frequently Asked Questions (FAQs), technical support requests, online discussion groups, Microchip consultant program member listing
- **Business of Microchip** – Product selector and ordering guides, latest Microchip press releases, listing of seminars and events, listings of Microchip sales offices, distributors and factory representatives

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The Development Systems product group categories are:

- **Compilers** – The latest information on Microchip C compilers, assemblers, linkers and other language tools. These include all MPLAB C compilers; all MPLAB assemblers (including MPASM assembler); all MPLAB linkers (including MPLINK object linker); and all MPLAB librarians (including MPLIB object librarian).
- **Emulators** – The latest information on Microchip in-circuit emulators.This includes the MPLAB REAL ICE and MPLAB ICE 2000 in-circuit emulators.
- **In-Circuit Debuggers** – The latest information on the Microchip in-circuit debuggers. This includes MPLAB ICD 3 in-circuit debuggers and PICkit 3 debug express.
- **MPLAB IDE** – The latest information on Microchip MPLAB IDE, the Windows Integrated Development Environment for development systems tools. This list is focused on the MPLAB IDE, MPLAB IDE Project Manager, MPLAB Editor and MPLAB SIM simulator, as well as general editing and debugging features.
- **Programmers** – The latest information on Microchip programmers. These include production programmers such as MPLAB REAL ICE in-circuit emulator, MPLAB ICD 3 in-circuit debugger and MPLAB PM3 device programmers. Also included are nonproduction development programmers such as PICSTART Plus and PIC-kit 2 and 3.

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- Distributor or Representative
- Local Sales Office
- Field Application Engineer (FAE)
- Technical Support

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Technical support is available through the web site at: http://www.microchip.com/support

DOCUMENT REVISION HISTORY

Revisions	Section/Figure/Entry	Correction
DS50003657B (04-04-24)	**Section 5.3.2.2 "Integrated Magnetics – ICM"**	Revised section title.
	Table 5-11	Revised information related to RGMII0 and RGMII1.
	Table 5-12	Changed GPIO 42 to 53 from "RGMII0" to "RGMII1."
	Section A.2.3 "Layer 3 – Signal Plane 1" and Figure A-4	Added "Signal Plane 1" to the section and figure titles.
	Section A.2.3 "Layer 3 – Signal Plane 1" and Figure A-5	Added "Power Plane 1" to the section and figure titles.
	Section A.2.8 "Layer 8 – Bottom Plane"	Added "Bottom Plane" to the section title.
	Section 5.3.1 "Network Ports"	Changed "RGMII0" to "RGMII1."
	All	Changed all Microchip part number "ZL80732" to "ZL30732B."
DS50003657A (02-23-24)	Initial release	

Chapter 1. Overview

1.1 INTRODUCTION

This hardware manual describes the design of the EVB-LAN9696 Reference Board, demonstrating the LAN9696RED Ethernet Switch device with PRP/HSR redundancy. The LAN9696 has a total bridging bandwidth of 66 Gbit/s.

Regarding basic board design, the LAN9696 and other derivatives, such as the LAN9694 and LAN9698, are all very similar devices. They use the same BGA package, pin layout, number of SerDes macros, and types. They also incorporate the VCore-IV CPU sub-system and associated interfaces. The main difference is the maximum bandwidth supported.

1.2 AUDIENCE

This document is developed primarily for hardware and software engineers who want to get an overview of designing products based on the LAN9696RED or its derivatives.

1.3 REFERENCES

1.3.1 Microchip Documents

Concepts and materials available in the following references may be helpful when reading this document. Visit www.microchip.com for the latest documentation.

- *LAN9696-24port Reference Schematic Design*
- *LAN9696 Data Sheet*
- *LAN8814 1G QuadPHY Data Sheet*
- *EV14Y36A PoE Evaluation Board User's Guide*
- *ENT-AN1187 Using the Serial GPIO/LED Controller*

1.3.2 IEEE Standards

- IEEE802.1D, Media Access Control Bridges
- IEEE802.1Q, Virtual Bridged Local Area Networks
- IEEE802.3, CSMA/CD Access Method and Physical Layer Specification
- IEEE1588-2019, Precision Clock Synchronization Protocol

1.3.3 (Optical) Module Standards

- SFP+ MSA, ftp://ftp.seagate.com/sff/SFF-8431.PDF

1.4 ACRONYMS AND DEFINITIONS

TABLE 1-1: ACRONYMS AND DEFINITIONS

Term	Definition
AMS	Automatic Media-Sense
CLI	Command Line Interface
EMI	Electromagnetic Interference (emissions)
GUI	Graphical User Interface
ICM	RJ45 connector with Integrated Magnetics
JTAG	Joint Test Access Group (IEEE1149)
LVDS	Low-Voltage Differential Signaling
LVTTL	Low-Voltage TTL
NPI	Node Processor Interface
PCS	Physical Coding Sublayer
PHY	Physical layer device
PTP	Precision Time Protocol (IEEE1588)
SFP	Small Form-factor Pluggable transceiver
SFP+	Small Form-factor Pluggable transceiver for 10 Gbps
SI	Serial Interface (SPI)
SME	Small/Medium Enterprise
SSM	Synchronization Status Message
Sync-E	Synchronous Ethernet (ITU-T G.8262/Y.1362)
TF-A	Trusted Firmware for ARM
TWI	Two-wired Interface

Chapter 2. EVB-LAN9696 Reference Board

2.1 INTRODUCTION

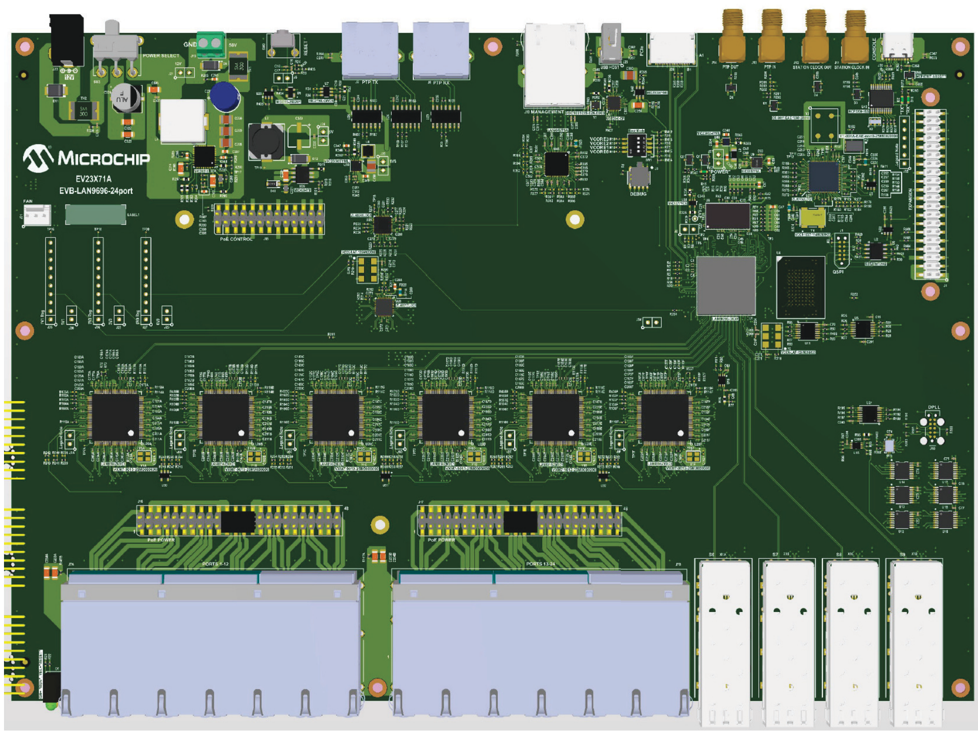
This section provides an overview of the EVB-LAN9696 Reference Board features, software, and hardware. See Figure 5-1 for details on the board's block diagram.

2.2 BOARD FEATURES

- One LAN9696RED switch
- Four SFP+ module slots (front), connected through SFI
- 24x tri-speed front ports using six LAN8814 Quad PHY's connected through QSGMII
- Network management port using LAN8840 single PHY connected through RGMII
- Serial UART port (via USB-C) for local management and software debugging
- USB Type A host port (for memory key - software updates)
- Sync-E DPLL ZL30732B to generate the board required clocks
- Two SMAs connectors for PTP and two for Station clock inputs and outputs
- Two ITU-T G.8275-compliant RS-422 interfaces for PTP applications
- Option for PoE add-on, like EV14Y36A (IEEE 802.3af/at/bt Type 4 standard compliant)
- Option for external CPU control via SPI and PCIe
- Expansion header – Raspberry Pi compatibility regarding power, UART, and I^2C.
- Barrel jack for 12V DC power and screw terminal for 48V/56V DC power.

2.3 BOARD DESIGN

FIGURE 2-1: EVB-LAN9696 REFERENCE BOARD



2.4 CPU SUB-SYSTEM

2.4.1 Embedded CORTEX-ARM53 CPU System

By default, the LAN9696 is managed by its embedded CPU system with on-board SDRAM, and Flash devices.

- Embedded ARM Cortex-A53 single core 64-bit processor operating at 1 GHz
- External memories include 1 GB DDR4 x16 RAM (Optional 2 GB can be mounted.)
- 2 MB QSPI NOR and 4 GB eMMC NAND (50 MHz DDR supported)
- Some GPIO's are implemented through the CPU sub-system serial GPIO engine (using inexpensive 74-series external shift registers) for connecting control signals to the four SFP+ slots and for LED control.

2.4.2 External CPU Connector

An external host CPU system can optionally be connected to control the LAN9696 Switch device through an on-board external CPU connector, using either:

- PCIe cable connector (Oculink)
- SPI interface located in the Expansion header (or in the QSPI Flash Program header)

The SPI client is set up through a specific boot mode. The PCIe end-point can either be set up through the SPI client or from a QSPI boot NOR Flash device.

2.5 MANAGEMENT

The embedded CPU system offers the Switch to be managed either through a Web GUI or through a Command Line Interface (CLI):

- Any Ethernet port connected to the Switch core can be used for local management through the Web GUI and/or Telnet/CLI session.
- A USB 2.0 port is on-board converted to the Switch serial UART port (Flexcom0) to allow CLI access using a standard terminal application, like PuTTY or TeraTerm.

2.6 COPPER PHYS

Ethernet ports are composed of the following tri-speed copper PHYs:

- LAN8814 Quad PHYs support Frame Pre-emption, time-stamping, and daisy-chaining of recovered line clocks for Sync-E applications.
- LAN8840 single-PHY supports Pre-emption and PHY time-stamping.

2.7 OPTIONAL ULTRA-POE FEATURE CONNECTOR

The 24 copper front ports support PoE/PoE+/ultra-PoE using a PoE add-on PCB. The PoE add-on performs all PSE functions, such as PD detection and classification, power feeding, and protection.

The PoE add-on connects to the magnetics' center taps (ICM type used), as well as to a low-bandwidth control interface (I^2C) connected to the embedded CPU sub-system.

PoE is managed as part of the general Switch management software.

The higher-voltage (48V~56V) power source for PoE is connected to the PoE add-on PCB. Therefore, the basic Switch reference board is not cluttered with PoE-related circuitry other than the passive feature connectors.

Presently, the newest version add-on PCB from Microchip uses the PD69210/PD69208T4 chipset, which supports both two-wired and four-wired ultra-PoE.

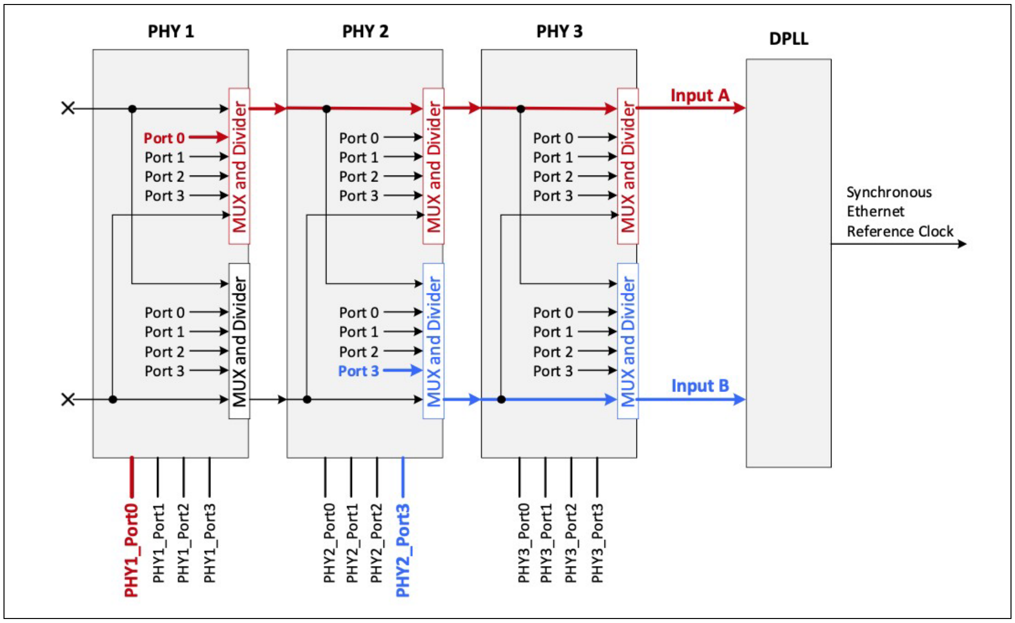
2.8 TIMING AND SYNCHRONIZATION

2.8.1 SyncE

The EVB-LAN9696 Reference Board supports SyncE through an on-board Microchip ZL30732B Sync-E controller, which is managed as part of the general Switch management software. This on-board DPLL generates all the reference clocks used by the Switch and PHY's. The clock source can be either a free-running oscillator or a clock recovered from one of the 10G ports on LAN9696 or from one of the CuPHY ports.

The board uses the LAN8814 CuPHY's capability to daisy-chain recovered PHY clocks and preselect which two clocks would be presented to the on-board DPLL as shown in Figure 2-2.

FIGURE 2-2: DAISY-CHAINING RECOVERED OUTPUT CLOCKS USING LAN8814



In the example, Synchronous Ethernet is configured to use recovered line timing from PHY1/Port0 as the primary reference, and the recovered line timing from PHY2/Port3 as the secondary reference.

The DPLL device prioritizes and selects the clock source under software control and attenuates jitter before presenting the resulting clocks for reference. A TCXO provides the base clock to the board DPLL, offering a Stratum-3 holdover. The TCXO can be removed for applications with less stringent requirements.

The board features an input and an output SMA for the DPLL device, supporting clock frequencies, such as 10 MHz, 2.048 MHz, and 1.544 MHz.

2.8.2 PTP/IEEE1588v2/802.1AS-2020

PTP interfacing is available through all network ports with high accuracy timestamping in the LAN9696, through ITU-T G.8275-compliant RS-422 1 PPS + ToD I/O ports (ePPS header), and through two separate 1PPS input and output SMA connectors.

The LAN9696 can be configured as boundary clock, transparent clock, PTP timeTransmitter, or PTP timeReceiver. The LAN9696 supports both one-step and two-step operations. Likewise, multiple time domains are also supported.

2.9 POWER

The reference board is powered through either a standard 12V DC barrel jack or a two-pin screw terminal for 48V/56V DC power input (for PoE and Industrial applications).

The 48V/56V DC power (15V-75V input range) is locally converted to 12V DC using a Step-Down Buck-regulator, MIC28515.

A two-position switch is used to select between the external 12V or the locally generated 12V power. The 12V power sources a number of local DC/DC converters, the FAN connector, and the PoE header.

Table 2-1 shows how the power supplies are generated and used on the board.

TABLE 2-1: DC/DC POWER SUPPLIES USAGE

Supply	DC/DC Converter	Used By
12V	0.9V by MIC24055.	LAN9696 Switch core and 10G SerDes I/O
	1.1V by MIC24055.	LAN8814/LAN8840 PHY core and analog low
	3.3V by MIC24055.	LAN8814/LAN8840, eMMC, ZL, USB, and SFP+
	5.0V by MIC4684.	USB and DC/DC
5V	1.2V by MIC23050.	LAN9696 VDDIO_DDR and DDR4 VDDQ
	1.8V by MIC23303.	LAN9696: VDDIO, VDDH, VDDPLLDDR ZL30732B: Core.
3.3V	2.5V by MIC5377.	DDR4

The expansion header can supply both 5V and 3.3V power.

NOTES:

Chapter 3. Management Software

3.1 INTRODUCTION

The EVB-LAN9696 Reference Board is managed remotely using a browser-based graphical user interface (Web GUI) or locally through a USB serial port supporting a command line interface (CLI).

3.2 CLI FOR MANAGEMENT AND DEBUGGING

The board can be connected directly to the USB port of a computer or laptop using a standard USB-to-USB Type C® cable to access the text-based CLI.

The Windows® driver and installer software file for the on-board UART converter is available at its homepage: (https://ww1.microchip.com/downloads/en/DeviceDoc/MCP2221_Windows_Driver_2021-02-22.zip).

When connected through the USB connector, the computer detects the on-board USB device, which includes giving the connection a COMx port number. From here, the port behaves like a standard serial port.

Once connected, point the Terminal program of choice to the new COM port, and set up the port to 115200 baud, 8 data bits, no parity, 1 stop bit, and no flow control.

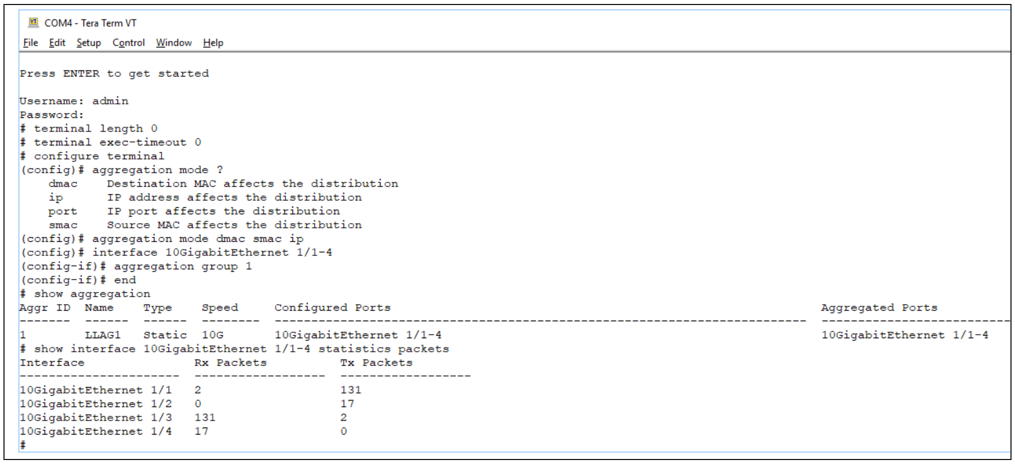
Log in to the Switch Application by using the default username, “admin” (without quotes), and leave the default password field blank. Help screens are available through the “?” or “help” command.

Note: To restore default settings and password, make a loop between ports 1 and 2 during power-on and allow the Switch Application to boot up completely.

One important use of the CLI is to determine the IP address of the switch when DHCP is enabled. This is done through the CLI command, “show interface vlan 1.” The command displays the IP address where the Web GUI is available.

An example of the CLI output when setting up a Static Aggregation on the 4x 10G ports is shown in Figure 3-1.

FIGURE 3-1: EXAMPLE CLI OUTPUT



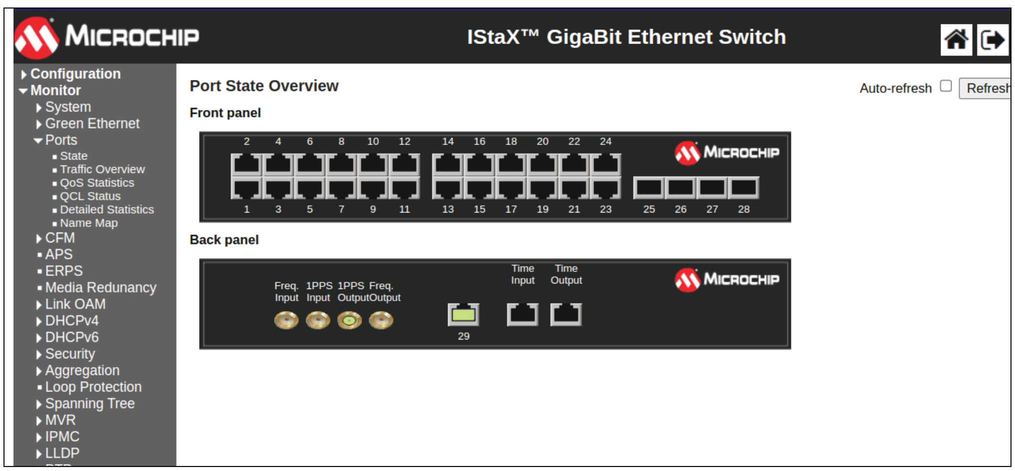
```
COM4 - Tera Term VT
File  Edit  Setup  Control  Window  Help

Press ENTER to get started

Username: admin
Password:
# terminal length 0
# terminal exec-timeout 0
# configure terminal
(config)# aggregation mode ?
    dmac    Destination MAC affects the distribution
    ip      IP address affects the distribution
    port    IP port affects the distribution
    smac    Source MAC affects the distribution
(config)# aggregation mode dmac smac ip
(config)# interface 10GigabitEthernet 1/1-4
(config-if)# aggregation group 1
(config-if)# end
# show aggregation
Aggr ID  Name    Type    Speed     Configured Ports                                                          Aggregated Ports
-------  ------  ------  --------  ------------------------------------------------------------------------  -------------------------
1        LLAG1   Static  10G       10GigabitEthernet 1/1-4                                                   10GigabitEthernet 1/1-4
# show interface 10GigabitEthernet 1/1-4 statistics packets
Interface               Rx Packets          Tx Packets
----------------------  ------------------  ------------------
10GigabitEthernet 1/1   2                   131
10GigabitEthernet 1/2   0                   17
10GigabitEthernet 1/3   131                 2
10GigabitEthernet 1/4   17                  0
#
```

3.3 WEB GUI FOR MANAGEMENT

The Web GUI is available from a computer or laptop connected to a network port. Ensure that the computer and switch are on the same IP subnet.

Launch a Web browser on the computer, enter the IP address of the switch (by default, the static IP address is 192.0.2.1), and enter the login credentials when prompted. The default username is “admin” (without quotes) and the default password field should be left blank. A graphical representation of the switch ports appears. See Figure 3-2 for the image of the Web GUI startup screen.

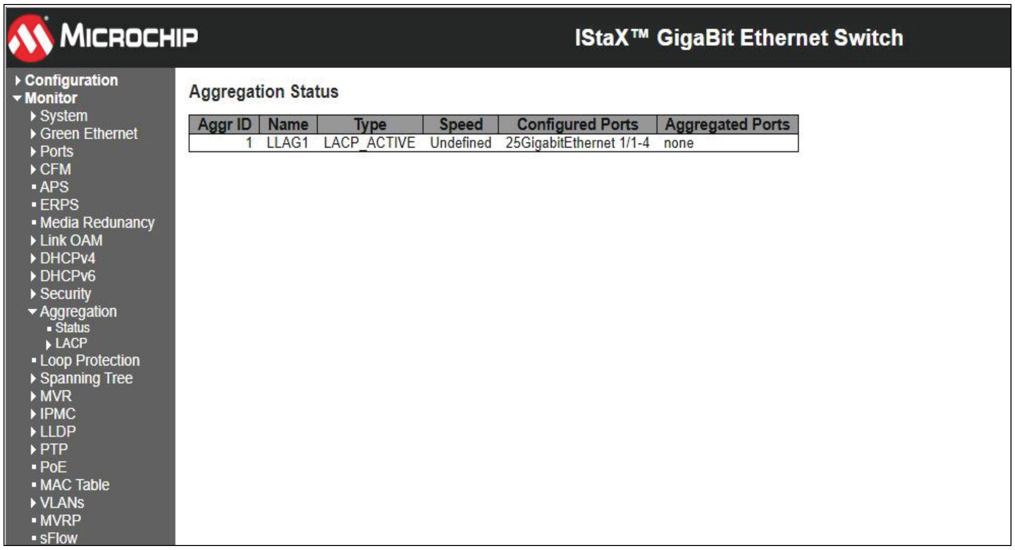
FIGURE 3-2: WEBGUI STARTUP SCREEN



Setup is done by selecting the **Configuration** tab at the top left and selecting an appropriate sub tab. Similarly, the status can be displayed through the **Monitor** tab. New versions of the Switch Application can be uploaded through the **Maintenance** tab. Help screens are available through the “?” button at top right.

Figure 3-3 shows an example Web GUI screenshot of the status of the previous Aggregation setup.

FIGURE 3-3: EXAMPLE WEB GUI SCREENSHOT

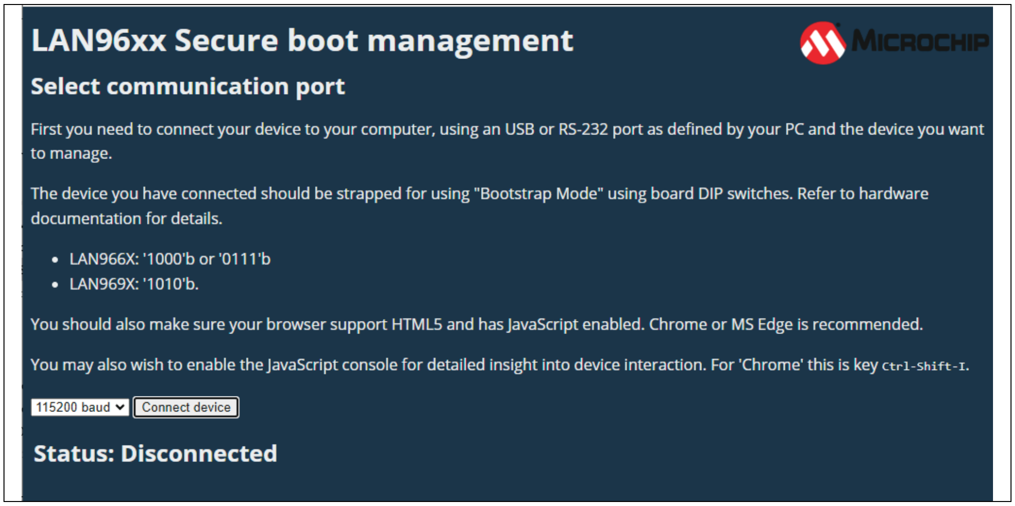


3.4 FIRMWARE UPDATE USING TF-A

A browser-based update utility called fwu.html allows a simple approach for recovering or updating the images on the EVB-LAN9696 Reference Board. Complete the following steps for a firmware update:

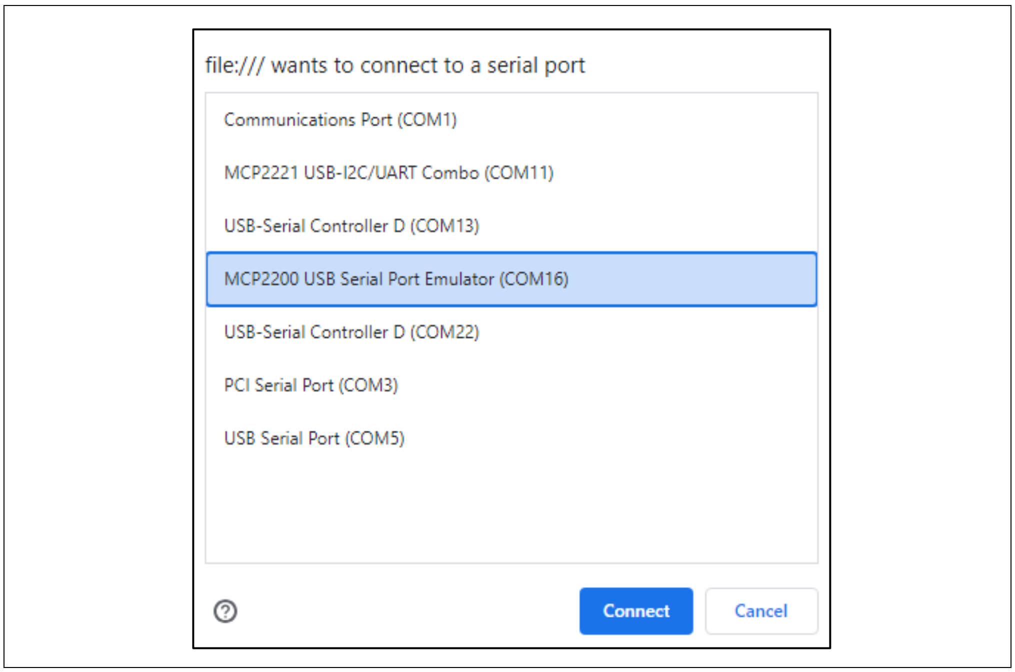
1. To have the fwu.html communicating with LAN9696, the board SW1 boot strapping must be set to ‘1010’ for 115200 baud or ‘1011’ for 921600 baud. (See Figure 3-4.) The built-in bootstrap manager, BL1 will enable the FLEXCOM0 UART interface (connected to the USB port) on the next boot-up.
2. To get the latest fwu.html file, go to the following URL and click on the **Latest** button found on the right side of the screen, under “Releases.”
   https://github.com/microchip-ung/arm-trusted-firmware.
   Download fwu-lan969x_a0-release.html and lan969x_a0-release-mmc.gpt.gz.
3. Make sure that the board USB port is not connected to a terminal server. If it is, then the HTML script will not work. A restart of the HTML script may be necessary.
4. Use either Chrome or MS Edge browser and have the browser setting, JavaScript, enabled. Simply run the fwu.html file inside the browser address field.
5. Click on the **Connect device** button. (See Figure 3-4.) The dialogue in Figure 3-5 appears.

FIGURE 3-4: CONNECTING TO LAN9696 FIRMWARE TOOL



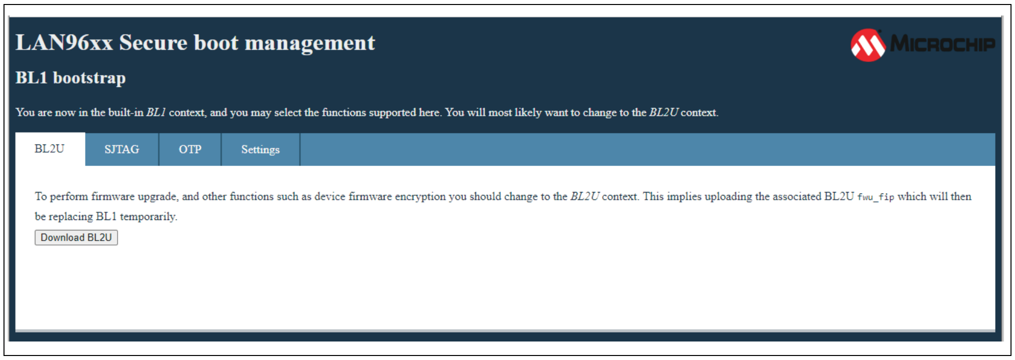
6. Select MCP2200 and click on **Connect**.

FIGURE 3-5: COM PORT SELECTION



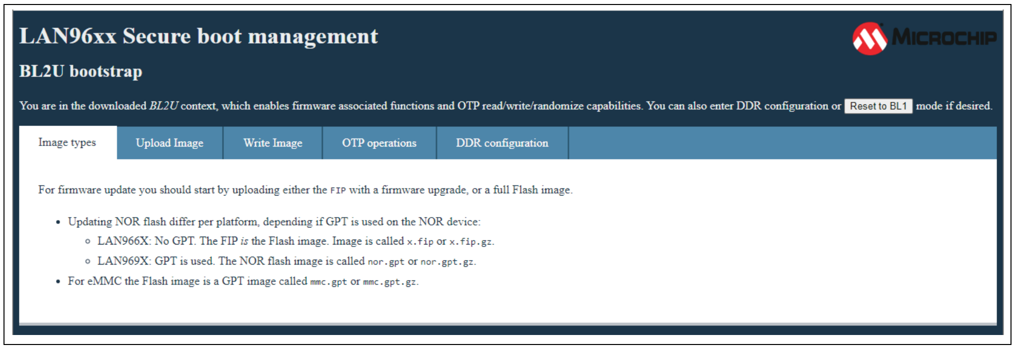
7. Figure 3-6 displays after clicking on **Connect**. Click on **Download BL2U** to get the firmware update functions.

FIGURE 3-6: BL1 BOOTSTRAP PAGE



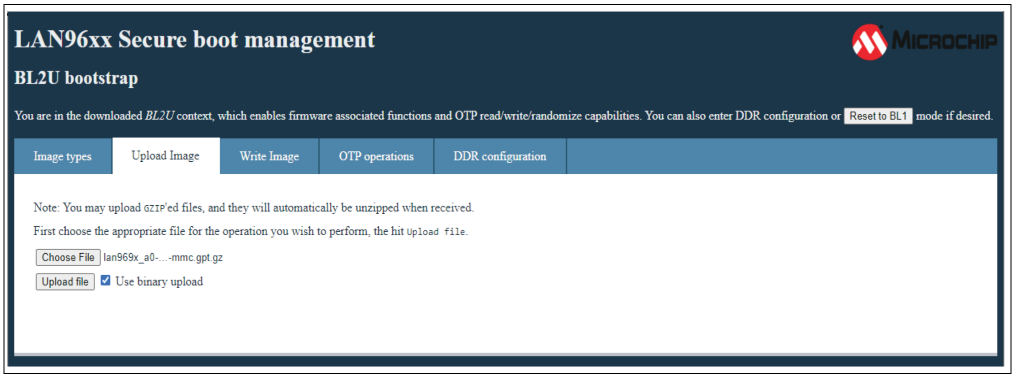
8. Within the BL2U monitor, select the **Upload Image** tab. See Figure 3-7.

FIGURE 3-7: BL2U FIRMWARE TOOLS



9. Click on **Choose File** and select the previously collected file lan969x_a0-release-mmc.gpt.gz. See Figure 3-8.

FIGURE 3-8: BL2U IMAGE UPLOAD

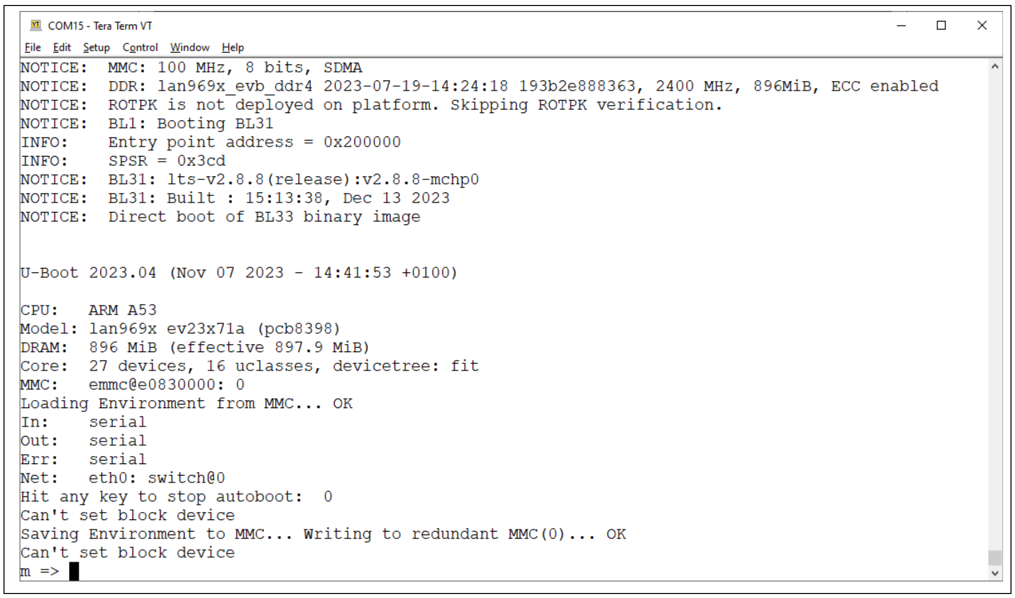


10. Click on **Upload file**. This will upload the file to the target RAM. It can take a while.
11. When done, select the next tab, **Write Image**, and choose eMMC as shown in Figure 3-9. Click on Write Flash Image. The Flash programming will take about 20 seconds.

FIGURE 3-9: BL2U FLASH PROGRAMMING

12. When the eMMC Flash programming is done, change SW1 DIP-switch back to '0000' and reset the board to boot up from the eMMC Flash.
13. Now running a terminal program on your computer connected to the USB port, the U-Boot prompt in Figure 3-10 appears.

FIGURE 3-10: INITIAL EMMC BOOT LOG



14. Recommended step: Run the following commands to have the U-Boot environment match the current U-Boot version:

```
m => env default -a
m => saveenv
m => reset
```

15. Format and upload image to the on-board eMMC typed NAND Flash.
    To store the Switch Application in the eMMC, ensuring faster boot-up time, the eMMC must be formatted as indicated below. The board will automatically get an IP address from the attached network DHCP server. Once done, the download of the ext4.gz file from the TFTP server begins.

```
ver=U-Boot 2023.04 (Nov 07 2023 - 14:41:53 +0100)

Environment size: 4022/65531 bytes
m => setenv mmc_fw 10.205.27.2:kbp/smb_lan969x.ext4.gz
m => run mmc_boot01_dlup
switch@0 Waiting for PHY auto negotiation to complete....... done
Port  29: Up (internal)
BOOTP broadcast 1
BOOTP broadcast 2
BOOTP broadcast 3
BOOTP broadcast 4
DHCP client bound to address 10.205.28.25 (3659 ms)
Using switch@0 device
TFTP from server 10.205.27.2; our IP address is 10.205.28.25;
sending through gateway 10.205.28.1
Filename 'kbp/smb_lan969x.ext4.gz'.
Load address: 0x64000000
Loading:
#################################################################
#################################################################
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###########################
         4.4 MiB/s
done
Bytes transferred = 19477211 (12932db hex)
Uncompressed size: 134217728 = 0x8000000
success!

MMC write: dev # 0, block # 534528, count 262144 ... 262144 blocks
written: OK
success!

MMC write: dev # 0, block # 2631680, count 262144 ... 262144
blocks written: OK
m =>
```

16. When finished, the SW1 DIP-switch should be set to ‘0000’ for eMMC boot-up. Reset the board to reboot, and the Linux image should start.

3.5 USING FIRMWARE UPDATE TO UPLOAD PCIE® DRIVER

The LAN9696 can be controlled externally by using it as a PCIe end point. The SerDes macro and PCIe controller are, however, not per default setup.

The Firmware update utility can be used to upload a special LAN969x_PCIe image to the NOR Flash. The image configures and enables the PCIe end-point interface. The bin file does not contain U-boot.

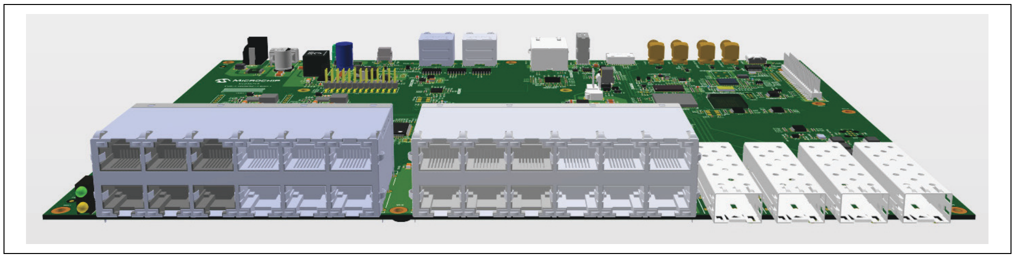
The LAN969x_PCIe image is released along with the TFA software.

Chapter 4. LED Indicators and Board Connector

4.1 FRONT BRACKET LAYOUT

Figure 4-1 shows the front bracket layout for the EVB-LAN9696 Reference Board.

FIGURE 4-1: FRONT BRACKET LAYOUT



4.1.1 Board Status and Reset LEDs

A double-LED (green/yellow, D1) is available to show the current board status. The green LED is controlled by the Switch Application, and the yellow LED is turned ON during board Reset.

4.1.2 Integrated Magnetics (ICM) and Port Status LEDs

The reference board uses two 2x6 RJ45 ICM with integrated LEDs to present the 24x tri-speed copper ports.

On each RJ45 connector, there are two LEDs available (green and yellow). The port status LEDs are automatically controlled by the LAN8814 PHY's. The left LED signals 1G link/traffic (green solid/blinks) and right LED signals 10/100M link/traffic (yellow solid/blinks).

4.1.3 SFP+ Slots

The reference board has four SFP+ slots for 10G transceivers. Each slot is organized as a single module and connects directly to LAN9696 10G SerDes macro, S6-S9.

The SerDes macro supports a variety of interface standards, such as SFI, 2500BASE-X, 1000Base-X, and 100BASE-FX to optical SFP modules, as well as copper SFPs through XFI/SGMII.

It can also support 10GBase-KR, 5GBASE-KR, 2500Base-KX, and 1000Base-KX by using DAC cabling.

Under each SFP slot, a single bi-color LED (green/red, D7) signals the current port status. The LED status indication is controlled through the Switch serial GPIO engine.

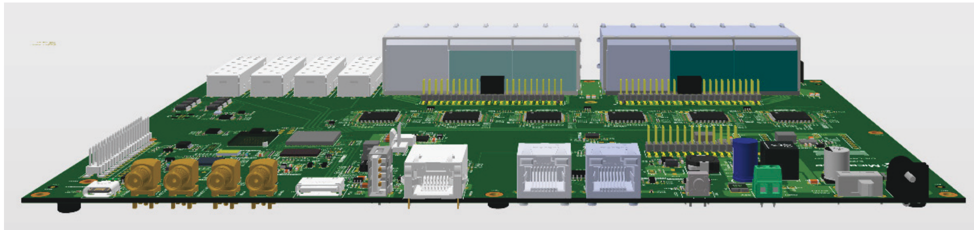
Likewise, the SFP MSA out-of-band control and monitoring of each SFP slot is made through the Switch serial GPIO engine.

The SFP I^2C required clock muxing is made by using an (external) 8-channel analog multiplexer. Three select pins are controlled through the Switch serial GPIO engine.

4.2 REAR-SIDE LAYOUT

Figure 4-2 shows the rear-side layout of the EVB-LAN9696 Reference Board.

FIGURE 4-2: REAR-SIDE LAYOUT



4.2.1 USB Type-C® Serial Port

The USB 2.0 Type-C port, J30, uses an on-board serial-to-USB converter MCP2200 from Microchip. The USB driver software (Windows®, Mac, Linux®) is found at:

https://www.microchip.com/wwwproducts/en/MCP2200

The USB converter has eight GPIOs, which can control board Reset and VCORE boot mode. Two GPIOs are controlling two data traffic LEDs: Tx/D2 and Rx/D16.

4.2.2 SMA Connectors

There are four SMA connectors. Two are used as inputs, and two are used as outputs. The SMA inputs are LVTTL and 3V3 tolerant—they are not 5V tolerant. The SMA outputs are also providing LVTTL levels.

1. **Station clock Input**, J11. Used for various input frequencies e.g., 1.544 MHz, 2.048 MHz, and 10 MHz to the board DPLL. The clock input is AC coupled.
2. **Station Clock Output**, J12. Used to provide various frequencies e.g., 1.544 MHz, 2.048 MHz, and 10 MHz from the board DPLL. The clock output is AC coupled.
3. **PTP IN**, J13. Used as 1PPS input into the board DPLL and the Switch PTP engine.
4. **PTP OUT**, J14. Used as 1PPS from either the board DPLL or the Switch PTP engine.

4.2.3 PCIe 2.0 – Oculink Connector

The LAN9696 implements a single-lane PCIe Gen2 end-point controller, that can be connected to any PCIe-capable system. The PCIe interface can be used by an external CPU to read and write the Switch registers.

The reference board offers PCIe connection over a standard Oculink cable connector, J28, when being set up as PCIe end-point.

4.2.4 ULPI USB 2.0 Host

USB 2.0 Host mode is supported by using a USB 2.0 female type-A connector, J29, which exposes the local USB 2.0 ULPI port on LAN9696. The ULPI port is connected through a Hi-Speed USB 2.0 transceiver PHY. An on-board Power Distribution MOSFET switch provides the 5V_host sourcing, limited to 600 mA.

The actual host capabilities depend on the Linux OS system support – though an USB memory key has been tested.

4.2.5 Management Port and Status LED

The Management port supports the normal tri-speed links (10/100/1000M). Despite its name, the port is treated as a normal switch port within the switch application. The Management port can optionally be held in Reset until its DPPL provided clock is valid.

The associated RJ45 ICM connector has two LEDs (green and yellow). These port status LEDs are automatically controlled by the LAN8840 CuPHY. The left LED signals 1G link/traffic (green solid/blinks) and right LED signals 10/100M link/traffic (yellow solid/blinks).

4.2.6 RS-422 Serial Time Synchronization Interface

A RS-422 serial port is available for synchronizing the PTP time of the system. It terminates into two RJ45 connectors: one PTP timeReceiver (input) and one PTP timeTransmitter (output).

In addition to the RS-422 serial port RX/TX signals for exchanging time information, the RJ45 connectors include signals for 1PPS input and output, and loopback within the PTP timeReceiver RJ45 connector.

The interface complies with the ITU-T G.8271/G.703 specification.

4.2.7 Reset

A Reset button is available on the reference board. When pressed, it drives the input of a voltage supervisor low, thus creating a hard Reset to the board. A yellow LED (at D1) is used for indication.

If the Reset button is held, Reset is released. The state of the Reset button can be read by the software once it is running again to determine ‘long press’ and reset the board back to the default setup. However, this is not currently supported by the Switch Application.

4.2.8 Power Select Switch

The slide switch, SW3 used for power selection between 12V DC (J23) or 48V/56V DC input (J19) has a 5A rating. When using the 12V DC input the maximum current is less than 4A.

A single green LED (at D15) indicates Power-ON using the 5V supply. This LED is controlled by a 1.2V power good signal.

4.3 ADDITIONAL ON-BOARD CONNECTORS

4.3.1 Expansion Header

The Expansion header can give an external CPU control over the SPI client register access interface or it can be used for programming the on-board NOR Flash device. Likewise, the on-board Boot mode strapping of VCORE_[3:0] can be overruled.

The Expansion header is partially compatible with Raspberry Pi. GNSS modules (designed for RPI) can be mounted to provide GNSS time input to the system using UART and 1PPS signals.

The Expansion header exposes various LAN9696 GPIO signals in alternate mode as shown in Table 4-1.

TABLE 4-1: GPIO SIGNALS IN ALTERNATE MODE

Signal	Alternate Mode
I^2C	SCL, SDA (Host mode, FLEXCOM 3)
UART	RXD, TXD – RS-422 (Host mode, FLEXCOM 1)
PTP	1PPS_IN (PTP.SYNC5)
VCORE_[3:0]	Boot mode selection. SW1 must be set to ‘0000’.
SI	SPI.SCK, SPI.D1, SPI.D0, and SPI.nCS
Reset	Board Reset (Note that NOR Flash has no Reset)

4.3.2 Ultra PoE Connector

According to **Section 2.7 “Optional Ultra-PoE Feature Connector”**, the 24 copper ports can support PoE using a Microchip 24-port PoE add-on PCB. The PCB connects to the magnetics’ center taps, through J16 and J17, as well as to a low-bandwidth control interface, through J15, controlled by the Switch CPU sub-system.

The power to the center taps (i.e., to power remote devices) must be supplied by the PoE add-on PCB itself, which must have a local ~48V DC power input and a 2 kV isolation (e.g., for the CPU I^2C or SPI interfaces) located on the PoE add-on PCB. Most of the PoE specific circuitry is located on the add-on PCB, leaving the reference board free from any PoE specific circuitry.

4.3.3 ARM CPU JTAG Connector

The board has a standard JTAG/Debug ten-pin (0.05”) header, J3, which can be used for boundary scan and ICE. System default is selecting ICE mode through resistor strapping.

4.3.4 QSPI Flash Program Header

The NOR boot Flash is normally programmed on-board using an external Programmer through the Expansion pin header, J4. The Expansion header implements a simple level conversion by resistor dividing.

The board also offers a QSPI Flash Program header, J1 (footprint only). It is intended for use in production.

Both Flash devices can also be programmed using the BL2U via the console port.

4.3.5 DPLL Programming Header

The board provides a DPLL programming header, J10 (footprint only), which is used with the Microchip Azurite GUI for the ZL30732B.

4.3.6 On-die Temperature Sensor Header

Footprint for a two-pin header, J2, makes it possible to do measurements on the anode and cathode signals of the on-die thermal diode, if required. Optionally, a temperature monitor EMC1812T-A can be mounted.

The LAN9696 includes another internal thermal diode for measuring the junction temperature and an embedded A/D converter for monitoring it through the register interface.

4.3.7 FAN Connector

A FAN connector, J31, is available next to the left edge of an attached PoE add-on PCB.

The FAN connector connects to a single FAN controller embedded in LAN9696 through a simple gating circuit. This allows for the FAN's health to be monitored at a given time through a Tacho input.

The FAN connector is powered from the 12V supply through a PWM-controlled FET for adjustable FAN power by doing ground chopping.

4.4 BOOT MODE AND REFERENCE CLOCK

LAN9696 Boot mode is strapped through its **VCORE[3:0]** pins using a dip switch, SW3. System default is to boot from the on-board eMMC NAND Flash device.

Additionally, it is possible to strap the source to the core reference clock through **REF-CLK_SEL**. As a default, the differential 156.25 MHz clock input is selected.

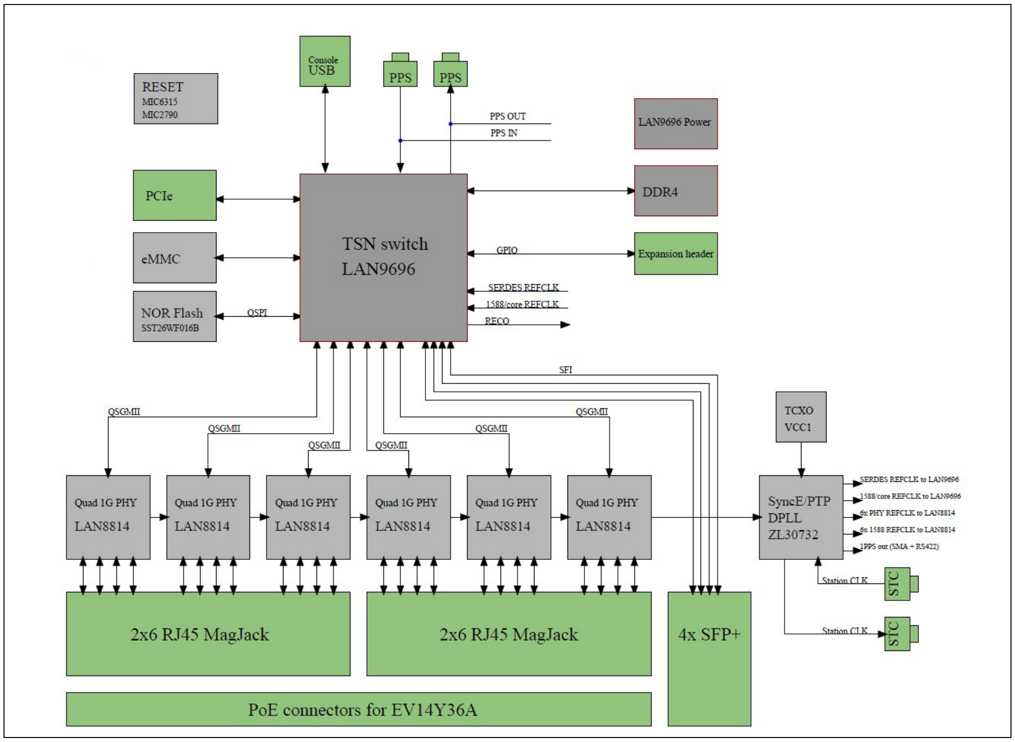
NOTES:

Chapter 5. Detailed Hardware Description

5.1 BLOCK DIAGRAM

Figure 5-1 depicts the block diagram of the EVB-LAN9696 Reference Board with its 24x tri-speed PoE capable copper ports and 4x SFP+ (10G) slots.

FIGURE 5-1: EVB-LAN9696 BLOCK DIAGRAM



The design is based on LAN9696RED Switch, which includes an ARM CPU, that connects to external DDR4 SDRAM and bootable Flash devices: QSPI NOR and eMMC NAND.

Local management and software debugging are done through a dedicated UART port or any Ethernet port. The LAN9696 UART port is connected to a serial-to-USB converter.

Optionally, the LAN9696 can be managed from an external CPU system connected through a PCIe 2.0 Oculink connector and through an SPI client found in the Expansion header, J4.

A Sync-E capable DPLL ZL30732B/Azurite generates the required reference clocks to LAN9696 and LAN8814 CuPHY's. The clock source can be either a free-running oscillator, TCXO type, a clock recovered from one of the 10G SerDes ports on LAN9696 or by one of the on-board CuPHY's. A recovered clock-chain links all the LAN8814 PHYs for pre-selection of the recovered clock source(s).

The DPLL jitter attenuates the selected clock source and provides it back to the Switch and PHYs where it is used as reference clocks. To avoid bit-wander, it is possible to source the generated clocks individually to make two different clock domains: A SerDes clock for Sync-E and a core clock for IEEE 1588.

There are four SMA connectors and two ITU-T G.8275-compliant RS-422 interfaces (1PPS + ToD input and output) which can be used for Sync-E and IEEE 1588 applications.

5.2 VCORE-IV CPU SUB-SYSTEM

The LAN9696 includes an internal CPU sub-system, called VCore-IV. This sub-system has an embedded ARM Cortex-A53 single-core, 64-bit processor operating at 1 GHz and various peripheral host controllers for DDR, SD/eMMC, MIIM, USB, and QSPI.

5.2.1 VCore Configuration Strapping

The LAN9696 has four dedicated strapping pins for selection of the initial Boot mode. These are named **VCORE[3:0]**. On the reference board, Boot mode strapping is made through a dip-switch, SW3.

Table 5-1 shows the possible boot modes that the reference board offers.

TABLE 5-1: BOOT MODE STRAPPING

VCORE[3:0]	Mode	Description
0000	eMMC FC0	Boot from eMMC Flash. Boot trace on Flexcom0 (115200/8/N).
0001	QSPI0 FC0	Boot from NOR Flash. Boot trace on Flexcom0 (115200/8/N).
0011	eMMC	Boot from eMMC Flash.
0100	QSPI0	Boot from NOR Flash.
1000	QSPI0 FC0 HS	Boot from NOR Flash. Boot trace on Flexcom0 (921600/8/N)
1010	TF-A FC0	TF-A monitor on Flexcom0 (115200baud/8/N)
1011	TF-A FC0 HS	TF-A monitor on Flexcom0 HS (921600baud/8/N)
1111	SPI client	QSPI0 is configured as SPI client. Internal CPU is disabled.

To let the ARM CPU boot from the eMMC interfaced NAND Flash device, the strapping must be set to ‘0000’ or ‘0011.’

The Flexcom0 UART interface is connected to the on-board serial-to-USB converter and is exposed on the USB Type-C port.

When the SPI client is selected, an external CPU can have access to the Switch register interface.

5.2.2 LCPLL Configuration Strapping

Through a resistor pin strap on REFCLK_SEL, the SerDes required 156.25 MHz differential clock is also used as base frequency for the LAN9696 switch core. However, it is possible by changing the strapping or from the software to change configuration to use the 25 MHz reference clock in IEEE 1588 applications to separate the Sync-E and PTP timing domains.

5.2.3 DDR4 SDRAM

The LAN9696 SDRAM interface is 16 bits wide and is targeted to operate at a clock rate of 1200 MHz, which therefore requires an SDRAM of speed grade 2400 Mtransfers/sec or better. (The reference design has been tested at 1333 MHz with good results.)

The reference board is equipped with a single 1 GB DDR4 SDRAM (x16), AS4C512M16D4-75BCN. A larger SDRAM of 2 GB can optionally be mounted instead.

The DDR IO supply and the DDR_VTT termination regulator, MIC5166, are sourced by 1V2. DDR_VREF is generated by using resistor divider on 1.2V. The MIC5166 can optionally provide it.

5.2.4 eMMC NAND Flash

The VCore-IV CPU sub-system can boot directly from an eMMC interfaced NAND Flash.

The reference board is equipped with 4 GB eMMC NAND, IS21ES04G-JQLI. Larger devices can be used, presently up to 64 GB. The NAND IO supply, VDDQ, is sourced with 1.8V to match LAN9696 IO.

5.2.5 QSPI NOR Flash

The reference board is equipped with 2 MB NOR QSPI boot Flash, SST26WF016B (SOIC-8). This Flash can operate both as normal SPI and as QSPI IO mode.

The NOR IO supply, VDD is sourced with 1.8V to match the dedicated QSPI interface on LAN9696.

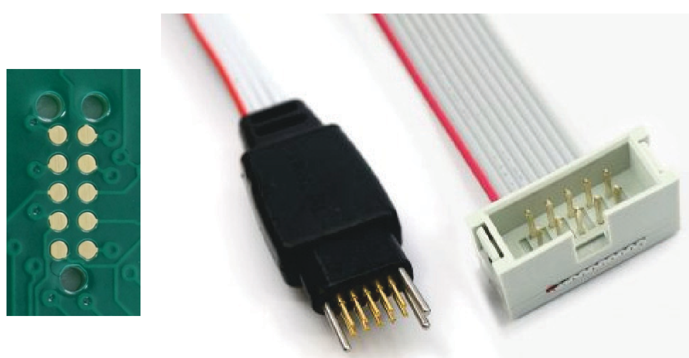
5.2.6 QSPI Flash Programming

The NOR Flash can be programmed using an external Programmer either through a 2x5 pin footprint header, J1, or through the Expansion header, J4, described in **Section 5.7 "Expansion Header"**.

Both headers include a Reset signal, through which the Programmer can keep LAN9696 in Reset, so it will release control of the SI Data output and therefore allowing only the Programmer to drive the signals while programming the QSPI NOR Flash.

Table 5-2 shows the footprint header and associated probe, which are intended to be used during production.

FIGURE 5-2: TC2020-NL-FP FOOTPRINT AND ASSOCIATED PROBE



The probe used with TC2050-NL-FP can be brought from Tag-connect.com homepage:

https://www.tag-connect.com/product/tc2050-nl-fp-footprint

Table 5-2 describes the location of the different signals in the 2x5 pin header, J1.

TABLE 5-2: QSPI NOR FLASH PROGRAMMING FOOTPRINT HEADER

Pin	Direction	Signal
1	In	QSPI.CK – SPI clock to SPI Flash
2	In	GND
3	BiDir	QSPI.D1 – SPI data to/from SPI Flash
4	In	1.8V Reset signal to Switch. Can be strapped to pin 6
5	BiDir	QSPI.D2 – SPI data to/from SPI Flash
6	Out	1.8V from board
7	In	QSPI.nCS – SPI chip select to SPI Flash
8	BiDir	QSPI.D3 – SPI data to/from SPI Flash
9	BiDir	QSPI.D0 – SPI data to/from SPI Flash
10	In	GND

5.2.7 ARM CPU JTAG Header

A standard ARM ten-pin header is used for boundary scan and ICE. It can be used with e.g., ARM DSTREAM on arm64. Table 5-3 describes the ten-pin header.

TABLE 5-3: ARM JTAG HEADER (TEN-PIN, PITCH 0.05”)

Pin	Direction	Signal	Description
1	Out	1V8 supply	3V3 supply
2	BiDir	TMS	JTAG Test Mode Select Input (See **Note 1**.)
3	In	GND	Ground
4	In	TCK	Test Clock Input
5	In	GND	Ground
6	Out	TDO	JTAG Test Data Output
7	—	BLIND	—
8	In	TDI	JTAG Test Data Input
9	In	nDETECT	Not used. 10 kΩ pull-up
10	In	nRSTD	JTAG_CPU_nRST – can be modified to be general Switch Reset, nRESET.

Note 1: Bidir in SingleWire Debug mode.

The LAN9696 provides a JTAG_SEL input signal used for selecting JTAG mode of operation. Table 5-4 shows which tap controller can be selected. Default mode is CPU Tap controller.

TABLE 5-4: JTAG MODE OF OPERATION

JTAG_SEL	Mode
0	Test Tap controller
1	CPU Tap controller

5.2.8 FLEXCOM Usage

LAN9696 VCore-IV CPU sub-system has four multi-purpose COM modules called FLEXCOM. The FLEXCOM[3:0] IOs are available as alternate functions on the GPIO pins.

5.2.8.1 UART FOR TF-A MONITOR AND CLI MANAGEMENT

The FLEXCOM0 UART is used as console port and is connected to an USB-to-UART serial converter, MCP2200.

The MCP2200 has 256-bytes user EEPROM and eight general purpose input/output pins. Four of the pins have alternate functions to indicate USB and communication status. The reference board uses the MCP2200 Tx/Rx functionality to indicate UART traffic through two green LEDs.

The MCP2200 must initially be programmed to set the on-chip signals GP[5:0] mode as input. Otherwise, factory default settings will keep holding the reference board in Reset.

Table 5-5 describes the MCP2200 GPIO usage.

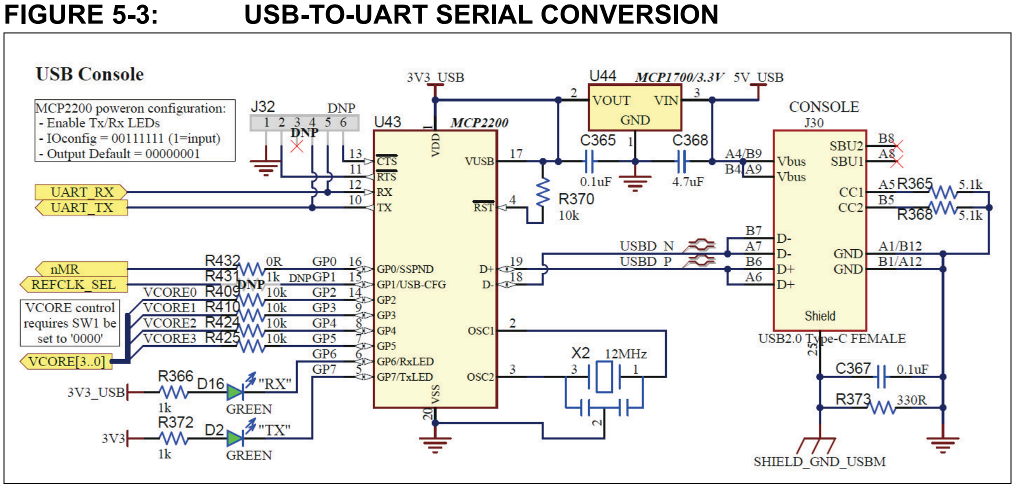
TABLE 5-5: MCP2200 GPIO USAGE

GP	Direction	Signal
0	Default set as input	nMR. Board Reset. (Default set to high.)
1	Default set as input	REFCLK_SEL. Core clock select, 0: 25 MHz, 1: 156.25 MHz.
2	Default set as input	VCORE0 (See **Note 1**.)
3	Default set as input	VCORE1
4	Default set as input	VCORE2
5	Default set as input	VCORE3
6	Out	RxLED functionality
7	Out	TxLED functionality

Note 1: SW3 must be set to ‘0000’ when GP2-5 are in use. Table 5-1 for Boot mode options.

Figure 5-3 shows the USB-to-UART Serial Conversion.

FIGURE 5-3: USB-TO-UART SERIAL CONVERSION



The USB converter is locally powered through its USB® Type-C connector using a low dropout (LDO) voltage regulator, MCP1700. It awakes smoothly, when connected to a Host USB.

5.2.8.2 UART FOR IEEE 1588 TIME INTERFACE – RJ45 G.703 PINOUT

FLEXCOM1 UART is connected to two RJ45-terminated RS422 connectors through ISL3180E transceivers. Over temperature, ISL3180E has well-defined propagation delays in both Rx and Tx directions, and the signal levels on the RJ45 pins are standard RS422 levels.

These two RJ45 time connectors, J8 and J9, expose both the UART signals used for time information and 1PPS PTP signals. The latter includes a 1PPS output, which is located in the timeTransmitter connector, and a 1PPS load/save input, which is located in the timeReceiver connector. A feedback path can be set up (PPS_EN0) by which, the PTP timeReceiver can loopback the 1PPS signal within the same RJ45 connector, hence allowing determination of the transmission delay of the RS422 transceivers and the UTP cable between two “time” connectors.

Table 5-6 and Table 5-7 describe the signals found in the two time connectors.

TABLE 5-6: TIMERECIEVER RJ45 CONNECTOR, J8

Pin	Direction	Signal
1	Out	1PPS FEEDBACK_TxN
2	Out	1PPS FEEDBACK_TxP
3	In	1PPS_RxN (load/save input). Local signal 1588 LD_SLV.
4	In	Ground
5	Out	10 kΩ pull-down to ground
6	In	1PPS_RxP (load/save input)
7	In	UART_RxN
8	In	UART_RxP

TABLE 5-7: TIMETRANSMITTER RJ45 CONNECTOR, J9

Pin	Direction	Signal
1	In	FEEDBACK_RxN. Local signal 1588 LD_MST.
2	In	FEEDBACK_RxP
3	Out	1PPS_TxN. Local signal PTP_OUT.
4	In	Ground
5	Out	10 kΩ pull-down to ground
6	Out	1PPS_TxP
7	Out	UART_TxN
8	Out	UART_TxP

> **Note:** The FLEXCOM1 UART signals are locally converted to 3.3V and can also be found in the Expansion header, J4.

5.2.8.3 SI INTERFACE FOR DPLL CONTROL

FLEXCOM2 SPI host interface is being used to control the on-board DPLL. This serial interface consists of the usual signals SPI clock DPLL.SCK (SCLK), data out DPLL.STX (MOSI), data in DPLL.SRX (MISO), and chipselect DPLL.nCS (nCS).

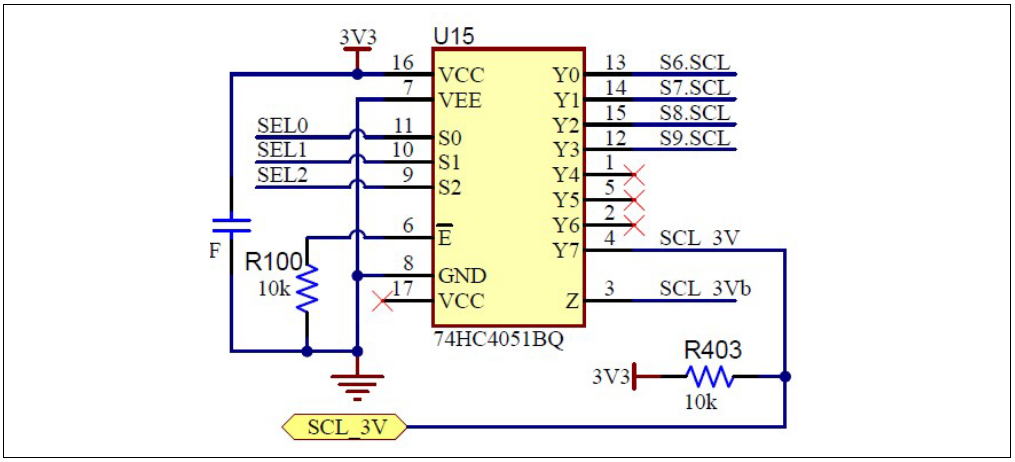
5.2.8.4 I²C INTERFACE

FLEXCOM3 TWIHS is being used to offer a general I²C interface for various usage: SFP, PoE, and to support devices connected on the Expansion header, J4. Before use, the two I²C signals, SCL and SDA, are voltage level converted to 3.3V by using a TI PCA9306.

SFP modules include an I²C EEPROM used for identification. However, all SFPs reside on the same I²C address, 1010_000, thus a multi-SFP system needs an I²C segment per SFP in order to avoid I²C address conflicts.

The LAN9696 has a built-in I²C multiplexer targeting this issue by offering ten gated clock signals. These separate TWI_GATE_SCL[9:0] lines are available on the Switch GPIO pins in alternate mode. However, the overall GPIO usage is constraint, and therefore the reference design implements the use of an external 8-channel analog multiplexer, 74HC4051BQ. See Figure 5-4.

FIGURE 5-4: 8-CHANNEL MULTIPLEXER



The three select pins, **SEL[2:0]**, are controlled through the Switch Serial GPIO engine.

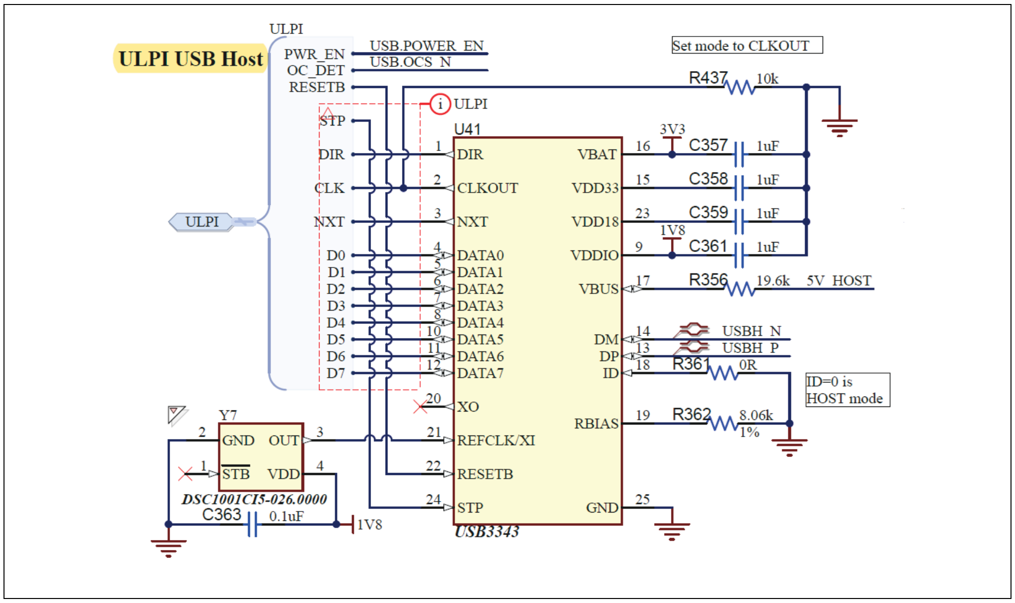
The four S[9:6].SCL signals are routed to the SFP+ slots on the front. The remaining clock signal, SCL_3V, is routed to the PoE Control header, J15, and the Expansion header, J4.

5.2.9 ULPI USB 2.0 Host

The LAN9696 built-in USB Host controller is connected to a Hi-Speed USB 2.0 transceiver PHY, USB3343 through its ULPI port interface located on the Switch GPIO in alternate mode.

Figure 5-5 shows how the external transceiver PHY has been implemented.

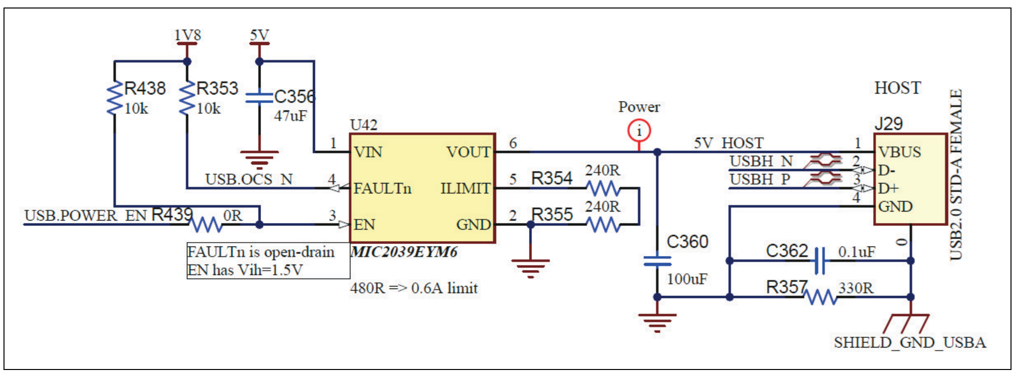
FIGURE 5-5: USB HOST PHY



USB 2.0 Host mode power sourcing is supported with a power distribution MOSFET, MIC2039. The external signals, POWER_EN (power enable) and OCS_N (overcurrent detection), are located on the Switch GPIO pins in alternate mode.

Figure 5-6 shows how the 5V (600 mA) power sourcing is provided to the USB 2.0 type-A connector, J29.

FIGURE 5-6: USB HOST POWER SOURCING



5.2.10 Serial GPIO Controller

The LAN9696 incorporates a Serial GPIO controller. It is documented in the *ENT-AN1187 Using the Serial GPIO/LED Controller*.

Through four pins, the controller can interface to 74-series shift registers on the board to provide a flexible GPIO mechanism. The four Serial GPIO pins are located on the Switch GPIO in alternate mode and named SG0_CLK, SG0_DO, SG0_DI, and SG0_LD.

Instead of using discrete shift registers, the external circuitry can be implemented in e.g., a central PLD. But the parallel inputs and outputs of this would then have to be routed across the board between the PLD and individual endpoints. By using shift registers, only the four LAN9696 Serial GPIO signals need to be routed “long distance” as each shift register can be distributed across the board as applicable.

In the reference design, the Serial GPIO controller is configured to use four bits per port and running 5 MHz. Its main task is to handle the out-of-band control signals towards the SFP slots, PoE connector, and RS-422 Time interface. Table 5-8 shows the Serial GPIO port usage.

TABLE 5-8: SERIAL GPIO CONTROLLER PORT USAGE

SG p[n]	Description
0-1	General IO. PoE control and I^2C clock selection
2-5	Disabled
6-9	SFP out-of-band control and port LEDs
10-29	Disabled

The SFP MSA defined signals, Tx Fault, Module Detect, and Rx Loss-of-Signal (LOS), are inputs and read through 74HC165 shift registers. Each of these signals has a 10 kΩ pull-up.

The SFP Rx LOS signals are each assigned to p[n]b[0] Serial GPIO input, which allows for a default connection within LAN9696 to the port PCS layer as Signal Detect input.

Likewise, the output signals, Tx Disable, Rate Select, LED1 (green, indicating full speed), and LED2 (yellow, indicating low speed), are individually controlled through another set of shift registers using 74HC594.

> **Note:** As the shift register outputs are set low by the board Reset, additional inverters may be required to set TX Disable high (active) during Reset. This can be done using inverting buffers, 74HC240. Likewise, the need for using pull-up resistors can be avoided.

Table 5-9 describes where the additional GPIO signals are located in the Serial GPIO engine.

TABLE 5-9: SERIAL GPIO CONTROLLER PIN DESCRIPTION

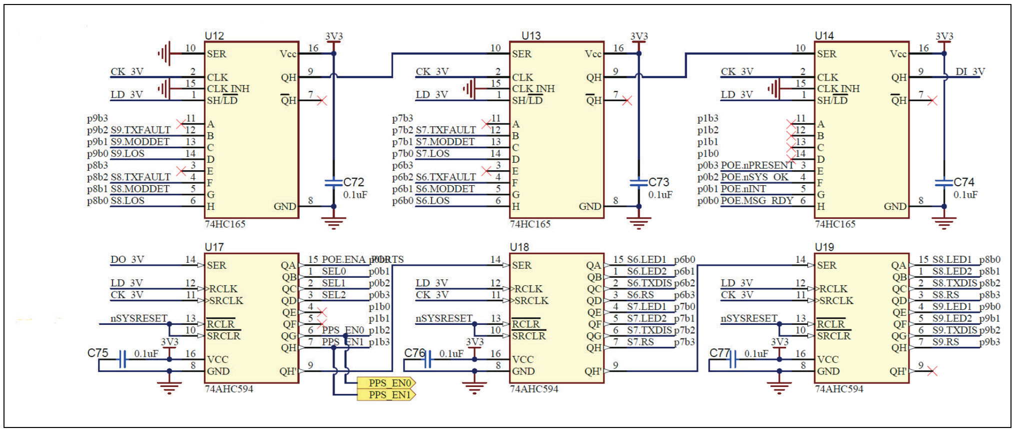
Port/Bit	Input	Output
p0b0	POE.nPRESENT	POE.ENABLE
p0b1	POE.nSYS_OK	SEL0 (I2C_SCL slot select)
p0b2	POE.nINT	SEL1
P0b3	POE.MSG_RDY	SEL2
p1b0	—	—
p1b1	—	—
p1b2	—	PPS_EN0 (RS422/1PPS Rx enable)
p1b3	—	PPS_EN1 (RS422/1PPS Tx enable)
p6b0	S6.LOS (1=LOS from SFP)	S6.LED1 – Green
p6b1	S6.MODDET (0=SFP present)	S6.LED2 – Red
p6b2	S6.TXFAULT (1=TX fault on SFP)	S6.TXDIS
P6b3	—	S6.RATESEL (RS1 and RS2)
p7b0	S7.LOS (1=LOS from SFP)	S7.LED1 – Green
p7b1	S7.MODDET (0=SFP present)	S7.LED2 – Red
p7b2	S7.TXFAULT (1=TX fault on SFP)	S7.TXDIS
p7b3	—	S7.RATESEL (RS1 and RS2)
p8b0	S8.LOS (1=LOS from SFP)	S8.LED1 – Green
p8b1	S8.MODDET (0=SFP present)	S8.LED2 – Red
p8b2	S8.TXFAULT (1=TX fault on SFP)	S8.TXDIS

TABLE 5-9: SERIAL GPIO CONTROLLER PIN DESCRIPTION (CONTINUED)

Port/Bit	Input	Output
p8b3	—	S8.RATESEL (RS1 and RS2)
p9b0	S9.LOS (1=LOS from SFP)	S9.LED1 – Green
p9b1	S9.MODDET (0=SFP present)	S9.LED2 – Red
p9b2	S9.TXFAULT (1=TX fault on SFP)	S9.TXDIS
p9b3	—	S9.RATESEL (RS1 and RS2)

Figure 5-7 shows the reference board Serial GPIO implementation using shift registers.

FIGURE 5-7: SERIAL GPIO SHIFT REGISTERS



5.2.11 MII-Management Bus Interface

The LAN9696 has two MII-Management bus interfaces. Both are available on the Switch GPIO pins in alternate mode.

MIIM0 is daisy-chained to the back port LAN8840 PHY and each of the six LAN8814 PHY's covering front port 1-24. The MII-Management addresses are 3 and 4-27, respectively.

MIIM0 is terminated at the end using pull-up on MDIO and split end-termination on MDC. Drive strength for MDC is 3.

5.2.11.1 PHY INTERRUPTS

LAN9696 Interrupt Controller has six dedicated Interrupt pins (operating as either input or output)—all located on the Switch GPIO pins in alternate mode. Two pins are meant for MIIM use but has no direct connection to the MDIO Controller as such.

In the reference design, all CuPHY's share a common interrupt signal using the Switch GPIO input: MIIM0_IRQ.

When an interrupt occurs on a shared interrupt source, the Switch Application polls the possible sources for their current interrupt status through the MII-Management interface as it is otherwise not possible to distinguish which source port has activated the shared interrupt signal.

The PHY shared interrupt signal, MIIM0_IRQ, has a 10 kΩ pull-up.

5.2.12 Adjustable FAN Speed and RPM Monitoring

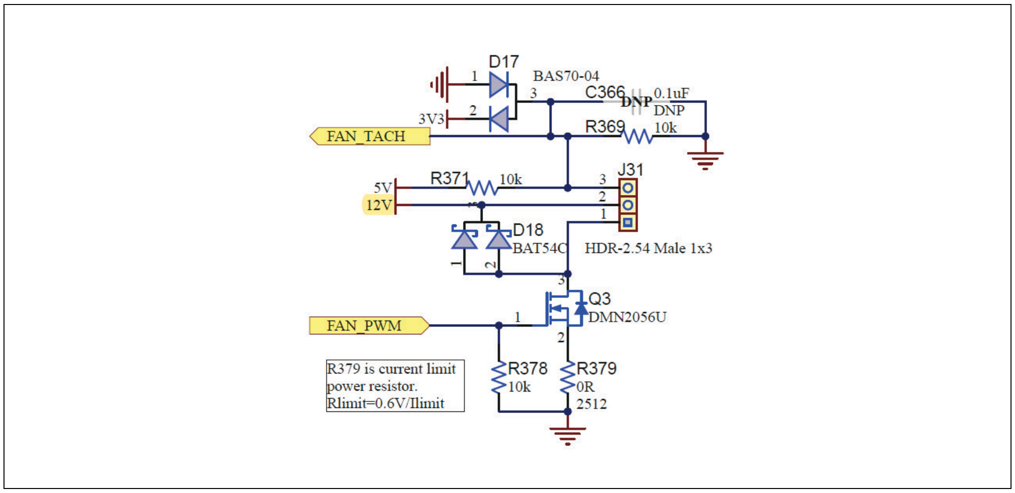
The reference board provides one FAN connector, which connects to the 12V power supply. By using LAN9696 built-in FAN controller, the FAN speed is adjustable through "ground chopping" with a PWM-controlled FET. FAN "health" is monitored through the LAN9696 Tacho input. Table 5-10 describes the FAN pin header.

TABLE 5-10: FAN PIN HEADER DESCRIPTION

Pin	Description
1	Tacho input. Set point to 2.5V.
2	12V output.
3	Return path. Ground chopped and current limited.

Both signals, PWM and TACHO, are located on the Switch GPIO pins in alternate mode. Figure 5-8 shows the FAN control implementation.

FIGURE 5-8: FAN CONTROL



It is possible to limit the current drawn by the FAN by changing resistor R379. Changing it to 6R would limit the current to 100 mA.

5.2.13 LAN9696 GPIO Usage

The previous sections have described the reference board usage of the alternate GPIO features. Table 5-10 summarizes the usage of all 67 GPIO pins in LAN9696. All GPIOs have internal pull-up.

TABLE 5-11: GPIO USAGE

GPIO	Mode	Signal	Description
0	ALT3	PCIe.PERST	PCIe Reset input
1	ALT3	USB.POW-ER_EN	USB Host Power enable output
2	GPIO	nBUTTON	Push button input. Long press detection
3	ALT1	FC0.RXD	FLEXCOM0 UART for TF-A and CLI.
4	ALT1	FC0.TXD	
5	ALT1	SGPIO.CK	Serial GPIO controller 0
6	ALT1	SGPIO.LD	
7	ALT1	SGPIO.DO	
8	ALT1	SGPIO.DI	
9	ALT1	MDC	MII-Management controller 0
10	ALT1	MDIO	
11	ALT1	MDINTn	
12	ALT3	USB.nRESET	USB Host PHY Reset output - RESETB

TABLE 5-11: GPIO USAGE (CONTINUED)

GPIO	Mode	Signal	Description
13	ALT3	USB.OC_DET	USB Host Over-Current Detect input
14	ALT1	eMMC.CMD	SD/eMMC Host controller 0
15	ALT1	eMMC.CLK	
16	ALT1	eMMC.D0	
17	ALT1	eMMC.D1	
18	ALT1	eMMC.D2	
19	ALT1	eMMC.D3	
20	ALT1	eMMC.D4	
21	ALT1	eMMC.D5	
22	ALT1	eMMC.D6	
23	ALT1	eMMC.D7	
24	ALT1	eMMC.RSTn	
25	ALT1	PWM	PWM output
26	ALT1	TACH	Tacho input
27	ALT1	SE.RC0	Sync-E Recovered Clock 0 output to DPLL
28	ALT1	FC1.RXD	FLEXCOM1 UART for RS-422
29	ALT1	FC1.TXD	
30	ALT2	ULPI.CLK	USB sub-system in Host mode
31	ALT2	ULPI.D0	
32	ALT2	ULPI.D1	
33	ALT2	ULPI.D2	
34	ALT2	ULPI.D3	
35	ALT2	ULPI.D4	
36	ALT2	ULPI.D5	
37	ALT2	ULPI.D6	
38	ALT2	ULPI.D7	
39	ALT2	ULPI.STP	
40	ALT2	ULPI.DIR	
41	ALT2	ULPI.NXT	
42	RGMII1	RX_D0	RGMII1 interface
43	RGMII1	RX_D1	
44	RGMII1	RX_D2	
45	RGMII1	RX_D3	
46	RGMII1	RX_CLK	
47	RGMII1	RX_CTL	
48	RGMII1	TX_D0	
49	RGMII1	TX_D1	
50	RGMII1	TX_D2	
51	RGMII1	TX_D3	
52	RGMII1	TX_CLK	
53	RGMII1	TX_CTL	
54	ALT1	SE.RC1	Sync-E Recovered Clock 1 output to DPLL
55	ALT2	FC3.SCL	FLEXCOM3 TWIHS – I^2C for various use
56	ALT2	FC3.SDA	
57	ALT4	PTP.SYNC3	1PPS output to PHY's

TABLE 5-11: GPIO USAGE (CONTINUED)

GPIO	Mode	Signal	Description
58	ALT4	PTP.SYNC4	1PPS output to SMA and RS-422
59	ALT4	PTP.SYNC5	1PPS input from SMA, RS-422, Expansion hdr
60	GPIO	nBRDRESET	Board Reset output
61	GPIO	STATUS	Status LED output – Green
62	GPIO	nPHYRESET	PHY Reset output
63	ALT2	DPLL.nCS	FLEXCOM2 using Flexcom_shared_17
64	ALT1	DPLL.SCK	
65	ALT1	DPLL.SRX	
66	ALT1	DPLL.STX	

5.3 SWITCH CORE

5.3.1 Network Ports

The EVB-LAN9696 Reference Board exposes one Management Cu-port in the back, and twenty-four Cu-ports and four SFP+ ports in the front.

The Management port uses a single CuPHY, LAN8840 connected to LAN9696 RGMII1 interface. The front Cu-ports use 6x QuadPHYs, LAN8814–each connected to LAN9696 SerDes interface using QSGMII. All PHY SerDes signals are AC coupled to ensure common-mode voltage compatibility.

Each QuadPHY has PHYAD[4:2] inputs for initial selection of the MII-Management bus addresses. The configuration of the PHYs is handled by the Switch Application. Details of the MII-management interface is described in **Section 5.2.11 "MII-Management Bus Interface"**.

TABLE 5-12: EVB-LAN9696 REFERENCE BOARD PORT MAPPING

Board Port	Internal Port	LAN9696 i/f	Description
Cu 1-4	D0-D3	S0	QSGMII, MIIM0 address 4-7
Cu 5-8	D4-D7	S1	QSGMII, MIIM0 address 8-11
Cu 9-12	D8-D11	S2	QSGMII, MIIM0 address 12-15
Cu 13-16	D12-D15	S3	QSGMII, MIIM0 address 16-19
Cu 17-20	D16-D19	S4	QSGMII, MIIM0 address 20-23
Cu 21-24	D20-D23	S5	QSGMII, MIIM0 address 24-27
SFP1	D24	S6	SFI
SFP2	D25	S7	SFI
SFP3	D26	S8	SFI
SFP4	D27	S9	SFI
—	D28	RGMII0	Unused
Management	D29	RGMII1	RGMII, MIIM0 address 3
PCIe	N/A	PCIE	PCIe end-point controller

5.3.2 Copper PHYs

5.3.2.1 MEDIA SIDE INTERFACE

The twisted pair interface on the copper PHYs are fully compliant with the IEEE802.3-2005 specification for CAT-5 media.

The PHYs integrate all passive components required to connect the PHYs' media interface to an external 1:1 transformer and common-mode choke. This reduces the number of components in a design and greatly simplifies the layout of this interface.

The PHYs support auto-negotiation and downshift and can automatically detect the speed of a link, if auto-negotiation is disabled, providing the appropriate connection. For the current tri-speed PHYs, this means 10M HDX/FDX, 100M HDX/FDX, or 1000M FDX.

The PHYs include Automatic Crossover Detection functionality for all speeds (HP Auto MDI/MDI-X function) and the ability to detect and correct polarity errors on all MDI pairs. These functions are normally enabled but can be disabled.

All PHYs support the IEEE standard range of 1m to 100m twisted pair cable. However:

- 1000Base-T mode requires Category 5 enhanced cable in accordance with the cabling specifications defined by IEEE802.3-2005.
- 100BASE-TX mode requires Category 5 cable, and 10BASE-T requires Category 3 cable as specified in ISO/IEC 11801.

5.3.2.2 INTEGRATED MAGNETICS – ICM

Instead of separate magnetic modules and RJ45 connectors, the reference board uses RJ45 connectors with integrated magnetics. This reduces the number of components on the board and the actual board area and allows for mounting options for PoE or non-PoE capable integrated magnetics whenever applicable.

The reference board uses two 2x6 ICM for the front copper ports. For use with the optional PoE add-on PCB detailed in **Section 2.7 "Optional Ultra-PoE Feature Connector"**, the Pulse J0B-3563NL and UDE RN6-1040F52F integrated magnetics can be mounted. For non-PoE capable applications, Pulse JC0-0132NL and UDE RM7-1040xxxF integrated magnetics can be mounted.

The Management port uses BEL L829-1J1T-43 ICM. Other pin-compatible parts may exist.

Generally, using good magnetics has a huge influence on EMI performance.

5.3.2.3 PORT LEDS

Each port has 2 LEDs, located in the RJ45 connector. One is green, and one is yellow LED. These are driven from the CuPHY's.

5.3.2.4 PHY GPIO USAGE

The following GPIO mapping is used on the LAN8814 PHYs.

TABLE 5-13: LAN8814 GPIO USAGE

GPIO	Signal	Description
0	only strapped	ALLPHY=0. Enable broadcast access on PHY address 0.
1	strap	MODE_SEL0=1, '11001' = ANEG and AMDIX.
2	1588_REF_CLK	DPLL clock output
3	1588_LD_ADJ	MODE_SEL1=0. 1PPS input from LAN9668 or DPLL.
4	only strapped	MODE_SEL2=0
5	only strapped	MODE_SEL3=1
6	only strapped	MODE_SEL4=1
7	RCVRD_CKI1x	RCVRD_CK_IN1
8	RCVRD_CKI2x	RCVRD_CK_IN2
9	RCVRD_CKO1x	RCVRD_CK_OUT1

TABLE 5-13: LAN8814 GPIO USAGE (CONTINUED)

GPIO	Signal	Description
10	RCVRD_CKO2x	RCVRD_CK_OUT2
11	P0.LED1	P0 port LED1. LED_MODE=1, individual control.
12	P0.LED2	P0 port LED2. PHYAD0=0.
13	P3.LED1	P3 port LED1. PHYAD1=0.
14	P3.LED2	P3 port LED2. PHYAD2.
15	PHYAD3	PHYAD3
16	PHYAD4	PHYAD4
17	P1_LED1	P1 port LED1
18	P1_LED2	P1 port LED2
19	P2_LED1	P2 port LED1
20	P2_LED2	P2 port LED2
21	—	Test point
22	COMA_MODE	Common COMA mode output. Pulled active during Reset. Connected to COMA input on all PHYs.
23	—	Test point

5.3.3 SFP+, SFI Interfaces

SFP signals are AC coupled in the SFP module and can therefore connect directly to the LAN9696 SFI high speed interfaces. The SFI interface can host a variety of optical modules by complying to different speeds: 10Gbit/s, 2.5Gbit/s, 1000BASE-X, and 100Base-FX, as well as copper SFPs through XFI and SGMII. It is also possible to support 10GBase-KR, 5GBASE-KR, 2500Base-KR, and 1000Base-KX by using DAC cabling.

The EVB-LAN9696 Reference Board serves four SFP+ slots using an on-board I^{2}C clock multiplexer. The semi-static control signals (RxLos, TxFault, TxDisable, Module-Detect, and RateSelect of the SFPs) are connected to LAN9696 Serial GPIO as detailed in Table 5-9. Likewise, the green and yellow LED signals to each SFP slot are also controlled by the Serial GPIO engine.

5.3.4 PCIe 2.0 End-point

Once enabled, the PCIe® Gen2 end-point controller gives access to the Switch registers. The PCIe end-point does support frame injection and extraction. The maximum possible bandwidth to obtain somehow depends on the ‘horsepower’ of the external CPU system.

Note:	The LAN9696 PCIe interface cannot operate as host/root-complex.

Table 5-14 describes the signals found in the on-board PCIe cable connector (Oculink). (BPTYPE=0, no I2C, PERST# (SB5) is open drain.)

TABLE 5-14: OCULINK PIN DESCRIPTION

Signal	Direction	Pin	Pin	Direction	Signal
3.3V act Rx	NC	A1	B1	NC	5V
GND	In	A2	B2	In	GND
PERp0/PCIE_RX_P	In	A3	B3	Out	PETp0/PCIE_TX_P
PERn0/PCIE_RX_N	In	A4	B4	Out	PETn0/PCIE_TX_N
GND	In	A5	B5	In	GND
PERp1	NC	A6	B6	NC	PETp1
PERn1	NC	A7	B7	NC	PETn1
GND	In	A8	B8	In	GND
2W-CLK	NC	A9	B9	In	BPTYPE/VSP, 10 kΩ pull-down
2W-DATA	NC	A10	B10	In	CWAKEn, OBFF/VSP, 10 kΩ pull-up
GND	In	A11	B11	In	GND
PCIE_PERSTn(VSP), 10 kΩ pull-up	—	A12	B12	In	PCIE_CLK_P/VSP
CPRESNT(VSP), 1 kΩ pull-down	—	A13	B13	In	PCIE_CLK_N/VSP
GND	In	A14	B14	In	GND
PERp2	NC	A15	B15	NC	PETp2
PERn2	NC	A16	B16	NC	PETn2
GND	In	A17	B17	In	GND
PERp3	NC	A18	B18	NC	PETp3
PERn3	NC	A19	B19	NC	PETn3
GND	In	A20	B20	In	GND
5V	NC	A21	B21	NC	3.3V act TX
MNT. SHIELD	—	0	—	—	—

5.4 RESET CIRCUITRY

The Reset circuitry consists of two Reset signals: nCLKRESET and nRESET.

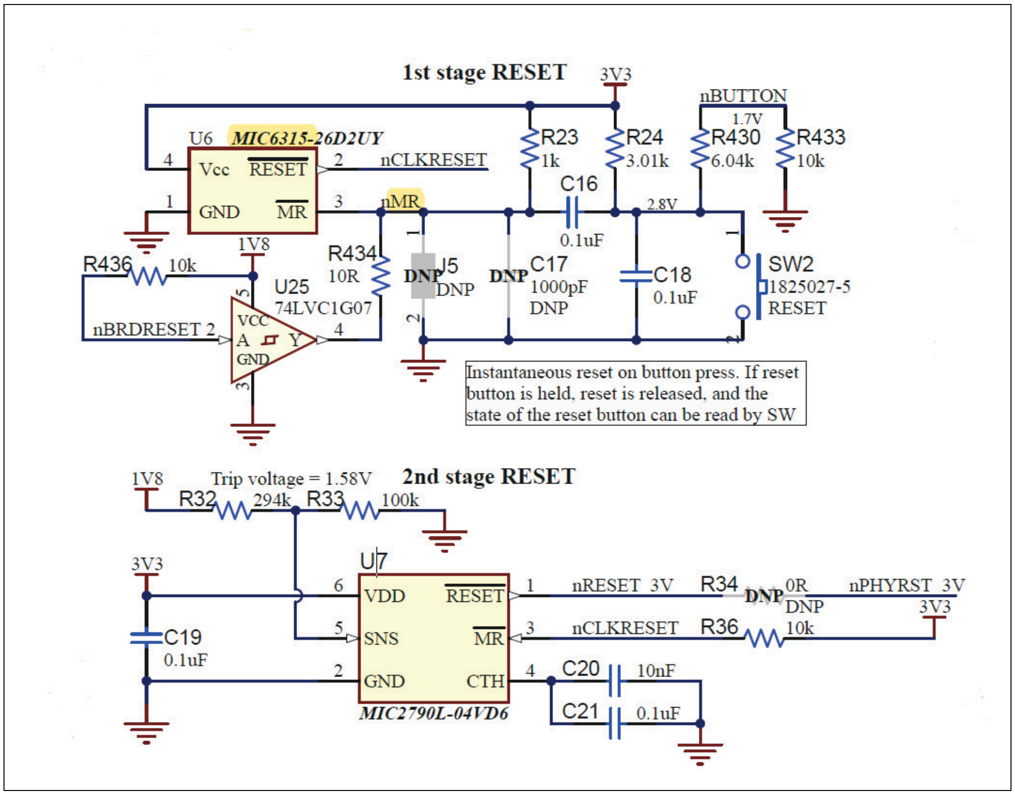
The nCLKRESET is used as master Reset to the on-board Sync-E DPLL. The nCLKRESET is generated by a Microchip MIC6315-26D2UY (2.6V, 20 ms) voltage supervisor, when the 3.3V supply falls below the threshold, or the Reset button (SW2) is being pressed.

The Reset button (SW2) is available on the back of the reference board. When pressed, it drives the input of the voltage supervisor low, creating a hard board Reset. It functions as a one shot, so pressing it causes a Reset pulse, but it (nBUTTON) can afterwards be sampled by firmware to determine if it is still being pressed during boot-up, causing a “Reset to factory default”.

The nRESET is the overall board Reset and is generated by using Microchip MIC2790L-04VD6 (0.4V) voltage supervisor with programmable Reset delay. nRESET_3V is asserted, when the sense voltage drops below the threshold, 0.4V, or when nCLKRESET is pulled low. The active-low Reset stays low for the duration of the Reset timeout delay once the sense voltage returns to normal and the manual Reset transitions to a Logic High state. The Reset timeout delay period is set to 200 ms using an external 100 nF capacitor on the CTH input pin. The voltage supervisor is monitoring the 1.8V supply with threshold is set to 1.58V.

Finally, board Reset can be generated or prolonged through nRESET_3V in the Expansion header, J4. See Figure 5-9.

FIGURE 5-9: RESET CIRCUITRY



During board Reset, a yellow LED, D1 is switched on. The PHY Reset can optionally be controlled by board Reset.

5.5 REFERENCE CLOCKS, SYNCHRONIZATION, AND PTP

5.5.1 Switch and CuPHY Reference Clocks

The reference board uses differential clock distribution to the Switch and CuPHY reference clock inputs. Differential distributed clocks offer the advantage of less jitter than single-ended clocks, which is required for complying with the reference clock specifications. Differential clocks (buffers and PCB traces) typically emit less radiated noise than single-ended clocks.

The LAN9696 is configured for a 156.25 MHz core reference clock input by pulling its **REFCLK_SEL** pin high. This is important for jitter performance reasons, when operating QSGMII and SFI interfaces. CuPHY's are configure for a 125 MHz reference clock by pulling their **REF_CLK_SEL** pins to '10'.

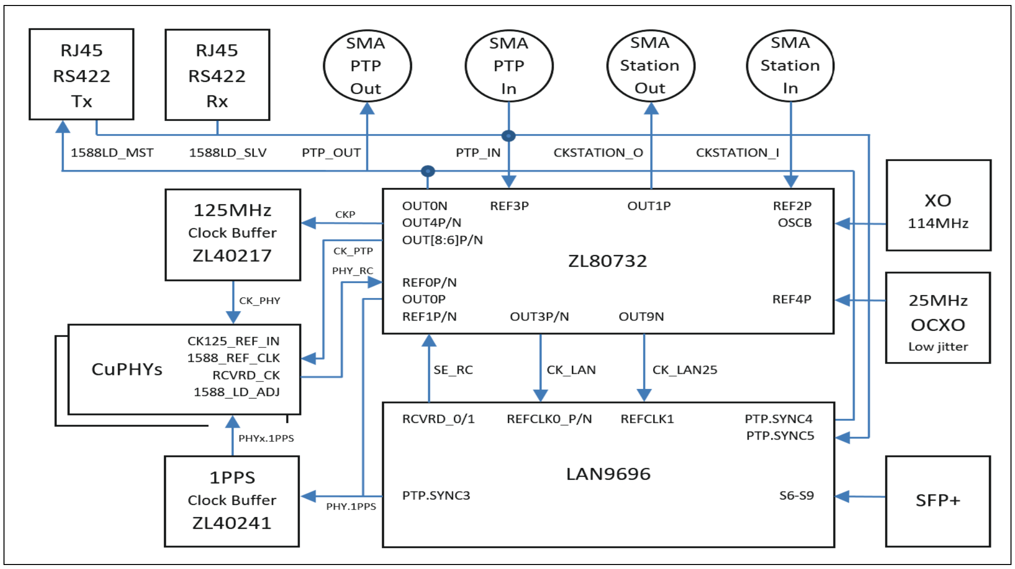
Recovered clocks, CuPHY 1588 and 1PPS clocks are distributed as single-ended clocks. A single-ended clock buffer, like ZL40241 typically offers a cost advantage compared to a differential clock buffer.

5.5.2 Sync-E and PTP Generated Reference Clocks

For Sync-E applications, all ports need to run from the same frequency source. Hence the reference board features central clocks that are distributed to the relevant recipients: Six QuadPHY's LAN8814 and one single PHY LAN8840 through a central clock buffer, ZL40217, and one directly to LAN9696.

The reference board uses a factory pre-programmed ZL30732B as frequency multiplier and clock buffer. The ZL30732B is operating in split-XO mode using a 114.285 MHz oscillator together with a 25 MHz TCXO for increased frequency stability. Using a TCXO or an OCXO with its superior frequency stability is a requirement for some SyncE and IEEE 1588 applications, but it can be omitted on applications not requiring this. Figure 5-10 provides an overall view of the clock distribution on the reference board.

FIGURE 5-10: REFERENCE CLOCKS, RECOVERED CLOCKS, SYNC-E, AND PTP DIAGRAM



A design based on the EVB-LAN9696 Reference Board, which does not require Sync-E and PTP capabilities, the ZL30732B can be replaced by the pin-compatible ZL30640, and the TCXO can be non-mounted.

The ZL30732B is programmed to generate the 156.25 MHz reference clock for LAN9696 REFCLK0 and a 25 MHz to LAN9696 REFCLK1 (optional use), and to provide a 125 MHz to the PHY clock buffer, ZL40217, and a 25 MHz LVCMOS to the 1588 1PPS clock buffer, ZL40241.

Table 5-15 shows ZL30732B frequency plan 00. The frequency plan is selected during Reset.

TABLE 5-15: ZL30732B FREQUENCY PLAN 'ZERO'

Signal	Description
OSCB	114.285 MHz clock input from XO
REF0P	25 MHz recovered clock from CuPHY (RCLK_PHY0)
REF0N	25 MHz recovered clock from CuPHY (RCLK_PHY1)
REF1P	25 MHz recovered clock from Switch (RCLK_SW/SE_RC0)
REF1N	25 MHz Sync-E feedback from OUT2N (RCLK_DPLL)
REF2P	10 MHz station clock input from SMA (CKSTATION_I)/J11

TABLE 5-15: ZL30732B FREQUENCY PLAN 'ZERO' (CONTINUED)

Signal	Description
REF2N	Optional IEEE 1588 feedback from OUT5P
REF3P	1PPS input from SMA (PTP_IN)/J13 or optionally from the time interfaces
REF3N	1PPS feedback from 1PPS clock buffer, ZL40241 (1PPS_FB)
REF4P	25 MHz from TCXO (for split-XO operation)
REF4N	NC (25 MHz)
Frequency plan selection and GPIO functionality	
AC[4:0]	Pull-down. AC = '00000'. Use frequency plan zero
DPLL0 is used for Sync-E. Drives Synth0 and Synth1.	
OUT1P	10 MHz station clock output to SMA (3.3V LVCMOS CKSTATION_O)/J12
OUT1N	TP9
OUT2P	25 MHz Management PHY reference clock (3.3V LVCMOS CK_RPHY25)
OUT2N	25 MHz Sync-E feedback from DPLL (3.3V LVCMOS RCLK_DPLL)
OUT3	156.25 MHz for Switch line reference clock (differential LVPECL CKL_P/N, 15 ms)
OUT4	125 MHz for QuadPHY line reference clock (differential LVPECL CKP_P/N)
DPLL1 is used for PTP. Drives Synth2.	
OUT0P	1PPS to PHY's and optionally Switch (1.8V LVCMOS 1PPS_1V8). Disabled.
OUT0N	1PPS to SMA, J14 and timeTransmitter, J9 (PTP_OUT converted to 3.3V). Disabled.
OUT5P	Optional IEEE 1588 feedback to REF2N
OUT6-8	25 MHz for QuadPHY 1588 reference clock (3.3V LVCMOS CK_PTP[5:0])
OUT9P	125 MHz for single-PHY 1588 reference clock (1.8V LVCMOS CK_PTP6)
OUT9N	25 MHz for Switch 1588 reference clock (1.8V LVCMOS CK_LAN25)

5.5.3 SMA Connectors

The reference board has two input and two output SMA connectors as detailed in Table 5-16.

One input SMA connector, STATION CLOCK IN, is used for various input frequencies e.g., 1.544, 2.048, and 10 MHz into the board DPLL.

The other input SMA connector, PTP IN is available for a 1PPS signal into the board DPLL and LAN9696 PTP Engine 5. Instead of coming from the SMA connector, the 1PPS input signal can also be sourced through a gated selection of the 1588 load/save signals from either the timeReceiver or timeTransmitter feedback in the two RS422 "Time" connectors. Their output enable is controlled by PPS_EN0 and PPS_EN1, respectively provided through the Serial GPIO engine.

One output SMA connector, PTP OUT can be controlled by either the board DPLL or LAN9696 PTP engine 4. The other output SMA connector, STATION CLK OUT, is solely controlled by the board DPLL. This clock is generated in the Sync-E clock domain. By default, the output frequency is 10 MHz. If other frequencies are wanted e.g., 1.544 or 2.048 MHz, this can be controlled by the running Switch Application.

TABLE 5-16: SMA CONNECTORS

SMA name	Connector	Description
STATION CLOCK IN	J11	SyncE clock to board DPLL/REF2P. AC coupled.
STATION CLOCK OUT	J12	SyncE clock from board DPLL/OUT1P. AC coupled.

TABLE 5-16: SMA CONNECTORS (CONTINUED)

SMA name	Connector	Description
PTP IN	J13	1PPS to board DPLL/REF3P and Switch PTP.SYNC5/GPIO59. Optional 1588LD_SLV or 1588LD_MST (feedback) signal.
PTP OUT	J14	1PPS from board DPLL/OUT0N or Switch PTP.SYNC4/GPIO 58.

The SMA inputs are LVTTL and 3.3V tolerant - *they are not 5V tolerant*. The SMA outputs also provide 3.3V levels.

5.6 ULTRA POE FEATURE CONNECTORS

The reference board has connectors for a 24-port Ultra-PoE/PoE+ add-on PCB.

The PoE add-on PCB connects to the magnetics' center taps on the reference board, as well as to a low-bandwidth control interface controlled by the Switch CPU sub-system. The power to the center taps (i.e., to power remote devices) must be supplied by the PoE add-on PCB itself, which must have local 48V/56V DC power input – and ensure 2 kV isolation on the CPU management interface.

For each group of 12 ports, the center tap connectivity is gather into one male connector as specified in Table 5-17.

TABLE 5-17: CENTER TAP CONNECTIVITY (MALE 2X24-PIN, PITCH 0.100”), J16/J17

Signal	Direction	Pin	Pin	Direction	Signal
VP_N0/N12	In	1	2	In	VP_N_U0/U12
VP_PMA	In	3	4	In	VP_PMA
VP_PMA	In	5	6	In	VP_PMA
VP_N1/N13	In	7	8	In	VP_N_U1/U13
VP_N2/N14	In	9	10	In	VP_N_U2/U14
VP_PMA	In	11	12	In	VP_PMA
VP_PMA	In	13	14	In	VP_PMA
VP_N3/N15	In	15	16	In	VP_N_U3/U15
VP_N4/N16	In	17	18	In	VP_N_U4/U16
VP_PMA	In	19	20	In	VP_PMA
VP_PMA	In	21	22	In	VP_PMA
VP_N5/N17	In	23	24	In	VP_N_U5/U17
VP_N6/N18	In	25	26	In	VP_N_U6/U18
VP_PMA	In	27	28	In	VP_PMA
VP_PMA	In	29	30	In	VP_PMA
VP_N7/N19	In	31	32	In	VP_N_U7/U19
VP_N8/N20	In	33	34	In	VP_N_U8/U20
VP_PMA	In	35	36	In	VP_PMA
VP_PMA	In	37	38	In	VP_PMA
VP_N9/N21	In	39	40	In	VP_N_U9/U21
VP_N10/N22	In	41	42	In	VP_N_U10/U22
VP_PMA	In	43	44	In	VP_PMA
VP_PMA	In	45	46	In	VP_PMA

Note 1: Maximum current VP_Nx is 1A.

TABLE 5-17: CENTER TAP CONNECTIVITY (MALE 2X24-PIN, PITCH 0.100”), J16/J17 (CONTINUED)

Signal	Direction	Pin	Pin	Direction	Signal
VP_N11/N23	In	47	48	In	VP_N_U11/U23

Note 1: Maximum current VP_Nx is 1A.

The Switch I^{2}C (FLEXCOM3 TWIHS) interface is provided in the PoE feature connector to manage the PoE devices located on the add-on PCB. See Table 5-18.

TABLE 5-18: POE MANAGEMENT INTERFACE (MALE 2X13-PIN, PITCH 0.100”), J15

Signal	Direction	Pin	Pin	Direction	Signal
PoE.nRESET	Out	1	2	In	PoE.nSYS_OK. 10k pull-up
PoE.ENA_PORTS	Out	3	4	In	GND
PoE.nPRESENT	In	5	6	BiDir	PoE.SDA – I^{2}C
PoE.nINT	In	7	8	Out	PoE.SCL – I^{2}C
GND	In	9	10	In	GND
SPI_MOSI/DO – no support	NC	11	12	NC	SPI_nCS – not supported
SPI_MISO/DI – no support	NC	13	14	NC	SPI_CLK – not supported
GND	In	15	16	In	GND
Spare	NC	17	18	In	PoE.MSG_RDY
Spare	NC	19	20	NC	Spare
GND	In	21	22	Out	12V
PoE.ID = 0	Out	23	24	Out	12V
3V3	Out	25	26	Out	12V

The control signals, PoE.nPRESENT, PoE.nSYS_OK, PoE.nINT, PoE.MSG_RDY, and PoE.ENA_PORTS, go through the serial GPIO engine. See **Section 5.2.10 “Serial GPIO Controller”**.

5.7 EXPANSION HEADER

The Expansion header, J4, is a 2x20, 0.1” pin connector and is partially Raspberry Pi compatible. It can be used for programming the NOR Flash or give an external CPU control over the SPI client register interface. It is also exposing various LAN9696 GPIO signals in alternate mode.

The GPIO mapping towards the Expansion header can be seen in Table 5-19. All signals are 3.3V levels.

TABLE 5-19: 2X20-PIN EXPANSION HEADER, J4

Signal	Direction	Pin	Pin	Direction	Signal
3.3V	Out	1	2	Out	5V
I2C SDA (FLEXCOM3)	BiDir	3	4	Out	5V
I2C SCL (gated SEL=7)	Out	5	6	In	GND
Spare	NC	7	8	Out	UART TXD (FLEXCOM1)
GND	In	9	10	In	UART RXD (FLEXCOM1)
Spare	NC	11	12	In	PTP_IN (SYNCE5) (1PPS)
Spare	NC	13	14	In	GND
Spare	NC	15	16	NC	Spare

TABLE 5-19: 2X20-PIN EXPANSION HEADER, J4 (CONTINUED)

Signal	Direction	Pin	Pin	Direction	Signal
3.3V	Out	17	18	In	VCORE2
Spare	NC	19	20	In	GND
Spare	NC	21	22	In	nRESET
Spare	NC	23	24	In	VCORE0
GND	In	25	26	In	VCORE1
Spare	NC	27	28	NC	Spare
Spare	NC	29	30	In	GND
Spare	NC	31	32	NC	Spare
VCORE3	In	33	34	In	GND
SPI.DI	In	35	36	In	SPI.nCS
Spare	NC	37	38	Out	SPI.D0
GND	In	39	40	In	SPI.SCK

Note: An add-on board using the Expansion header MUST have its own Power-On-Reset (POR), and the overall power consumption should be considered. Likewise, ensure that the VCORE[3:0] strapping settings are not overridden unintentionally. If the add-on board is intended to overwrite the board strapping, SW1 must be set to ‘0000’.

5.8 POWER DISTRIBUTION

The reference board is powered through either a standard 12V DC barrel jack, J23, or a two-pin screw terminal for 48V/56V DC power input, J19.

5.8.1 48/56V Input

The 48V/56V DC power (15V-75V input range) is locally converted to 12V DC using a step-down buck regulator, MIC28515. It can deliver a maximum of 5A and is set up to operate in HLL (Hyper Light Load) mode.

The MIC28515 offers a full suite of protection features to ensure safe operation of the device during Fault conditions. These include UVLO, external soft start to reduce current inrush, hiccup current limit short circuit protection and thermal shutdown.

5.8.2 DC/DC Converters

A slide switch, SW3 is used for power selection between 12V DC from jack or from the buck regulator. The 12V power supply sources a number of different 12V DC/DC converters:

- 0V9 for LAN9696 core supply (**VDD**) and SerDes10G IO (**VDDHS** and **VDDHV**). The DC/DC converter carries a calculated total load of 6.7A. Maximum current is 12A.
- 1V1(1.2V) for CuPHY’s core supply and analog low. New requirements for the LAN8814 core supply is to be above 1.15V. The calculated load is 5.0A. Maximum current is 6A.
- 3V3 for SFP and CuPHY’s IO and analog high. The calculated load is 4.8A. Maximum current is 6A.
- 5V for USB Host and additional DC/DC converters. The calculated load is 1.2A. Maximum current is 2A.

The 12V power supply also feeds direct power to the FAN connector and PoE Control header. Table 5-20 summarizes the 12V power usage.

TABLE 5-20: 12V DC POWER USAGE

12V DC/DC Converters	Supply (V)	Maximum Current (A)	Power (W)
MIC24055, LAN9696 Core, and 10G SerDes	0.9	6.7	6.0
MIC24055, LAN8814/8840 PHY Core, and Analog Low	1.1(1.2)	5.0	5.5
MIC24055, LAN8814/LAN8840 PHY IO, and SFP+	3.3	4.8	15.8
MIC4684, USB, and Additional DC/DC Conversion	5.0	1.2	6.1
Power Dissipation in Supply (@80% efficiency)	—	—	8.3
Direct Use: FAN and PoE Control Header	12.0	0.33	4.0
Total 12V	**12.0**	**3.8**	**45.6**

The MIC24055 DC/DC Converters are located on DC/DC modules. The generated 5V DC power supply sources local DC/DC converters:

- 1V2 for LAN9696 DDR4 (**VDDIODDR**) and DDR4L RAM (**VDDQ**). The calculated load is 480 mA. Maximum current is 600 mA.
- 1V8 for LAN9696 CMOS IO (**VDDIO18**), SerDes (**VDDH**) and DDR-PLL (**VDDPLLDDR**). The calculated load is 1.7A. Maximum current is 3A.

The 5V power supply also feeds direct power to FAN control, USB Host interface and the Expansion header. Table 5-21 summarizes the 5V power usage.

TABLE 5-21: 5V DC POWER USAGE

5V DC/DC Converters	Supply (V)	Maximum Current (A)	Power (W)
MIC23050, LAN9696 VDDIO DDR, and DDR4 VDDQ	1.2	0.5	0.6
MIC23303, LAN9696: VDDIO, VDDH, VDDPLLDDR ZL30732B: Core.	1.8	1.7	3.0
Power dissipation in supply (@80% efficiency)			0
FAN, Expansion header, and USB Host interface	5.0	0.5	2.5
Total 5V	**5.0**	**1.2**	**6.1**

A single green LED (D15) indicates Power-ON is using the 5V supply.

The generated 3.3V DC power supply sources most of the board components, like eMMC NAND Flash, OC/XO's, ZL devices, SFP slots, shift registers, and expansion header.

DDR4 VTT sink and source double data rate (DDR) termination regulator is sourced by 3V3. Table 5-21 summarizes the 3.3V power usage.

TABLE 5-22: 3.3V DC POWER USAGE

3.3V Devices	Supply (V)	Maximum Current (A)	Power (W)
eMMC NAND Flash	3.3	—	—
LAN8814/LAN8840 VDDIO and Analog High	3.3	2.3	7.6
ZL30732B and ZL40217 VDD. OCXO	3.3	0.3	1.0
USB Host PHY and Shift Registers	3.3	—	—
SFP+ Slot	3.3	1.8	6.0
Expansion Header and PoE Headers	3.3	—	—
Total 3.3V	**3.3**	**4.8**	**15.8**

5.8.2.1 POWER SUPPLY SEQUENCING

The strategy for the power supply sequencing is not to let a higher-powered supply feeding lower-level powered grids through the device internal protection diode network. The general power supply sequence is therefore:

- LAN9698: 0V9 -> 1V2 -> 1V8
- DDR: 1V2 -> 2V5
- Other (optional): 1V1 -> 3V3

The overall power sequencing is illustrated in Figure 5-11.

FIGURE 5-11: POWER SEQUENCING

```
  56V ─┬─ 12V ─┬─ 1V1 ─── 3V3
 Jack ─┘       ├─ 5V  ─┐        ┌─ 2V5
               └─ 0V9 ─┴─ 1V2 ─┴─ 1V8
```

5.8.2.2 POWER SUPPLY, TEST POINTS

A two-pin footprint can be found on each power supply output.

- 56V: J19
- 12V: J21
- 5V: J24
- 3V3: J22
- 2V5: TP21
- 1V8: J26
- 1V2: J27
- 1V1: J20
- 0V9: J25

Chapter 6. Environmental Requirements

6.1 STORAGE AND OPERATING CONDITIONS

The reference board is designed to operate within the following temperatures (case and airflow dependent):

- Operating temperature: 0° to 40°C
- Storage temperature: –10° to +70°C
- Operating humidity: 10% to 95% relative humidity, non-condensing

The reference board is provided without a case, so sufficient airflow might be needed, if the air temperature is above 30°C.

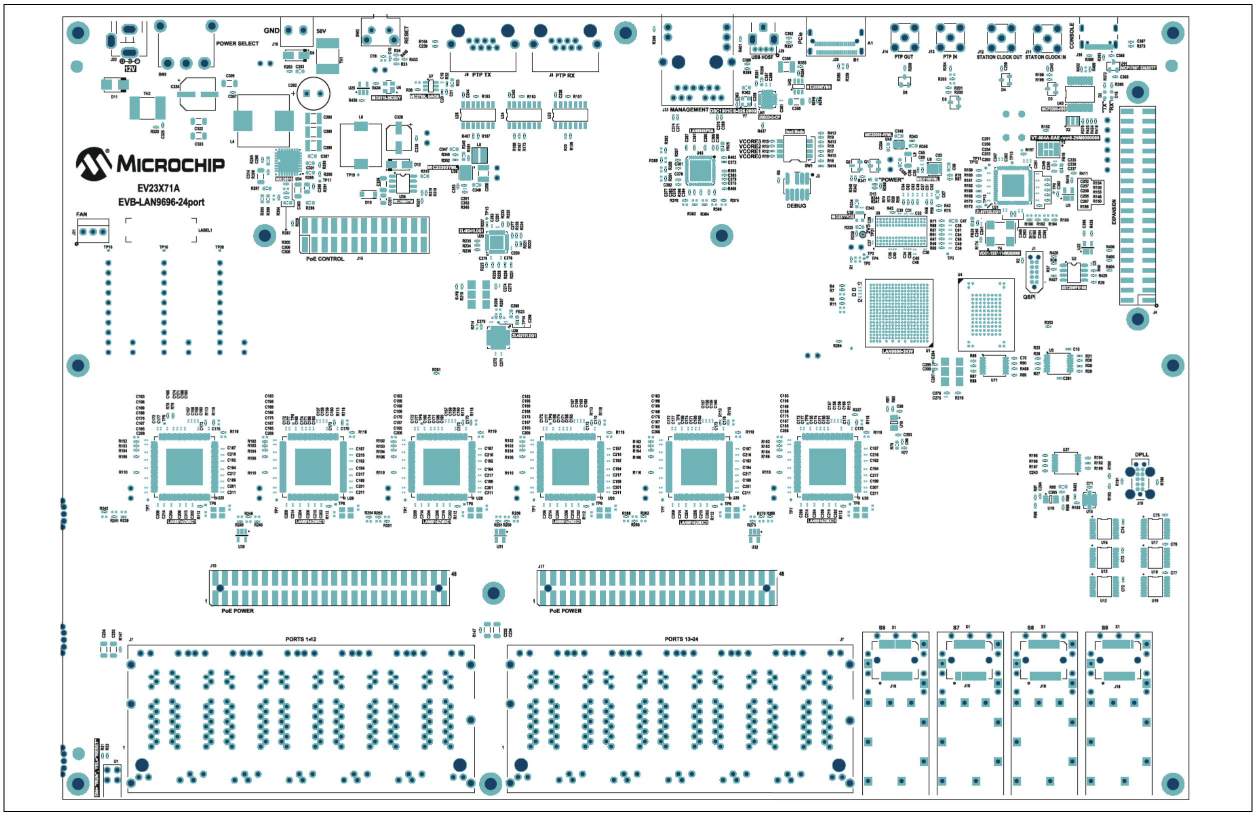
NOTES:

Appendix A. PCB Layout

A.1 DIMENSION

Figure A-1 shows the board outline. The board dimensions are 288 x 200 mm.

FIGURE A-1: EVB-LAN9696 REFERENCE BOARD OUTLINE



A.2 PCB LAYERS

The EVB-LAN9696 Reference Board uses eight PCB layers. (Refer to Table A-1.)

Solid ground planes on layers 2 and 7 ensure that all signals on layer 1 (top) and 8 (bottom) have a good reference plane. The two ground planes are also used to remove heat from components, and it must (also for this reason) be ensured that good connection between outer layer ground fills and the ground planes are established. (The outer layers do not have ground fill areas for copper balancing purposes.)

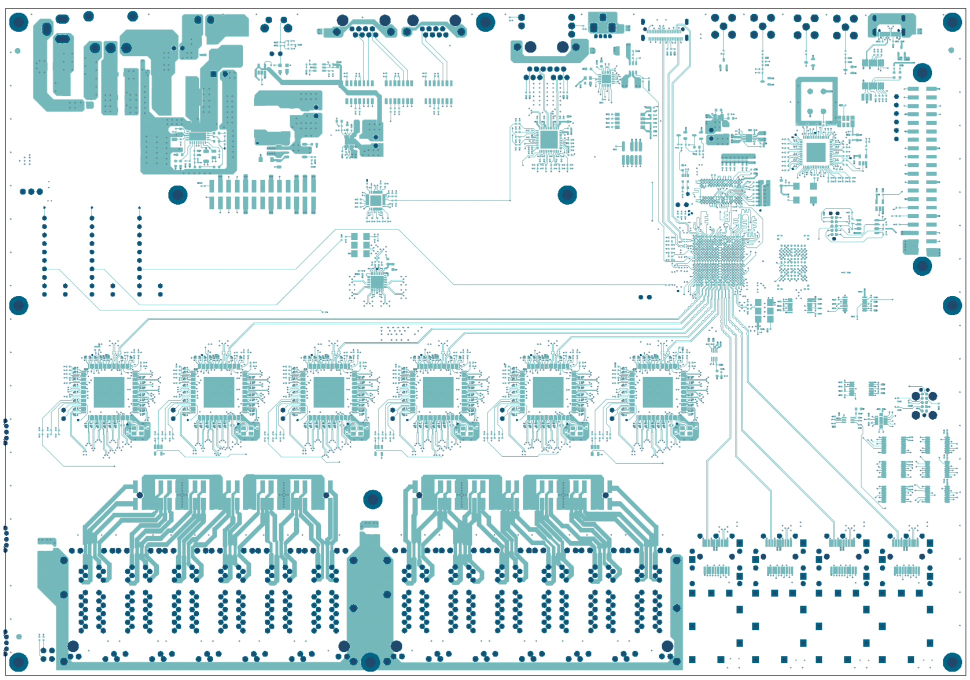
TABLE A-1: PCB LAYER OVERVIEW

Layer	Description
1 (Top)	LAN9696 signal traces for DDR4L. SerDes Tx paths to QuadPHy's and SFP.
2	Ground plane 1.
3	Signal plane 1. Signal traces for DDR4L. Various clocks for CuPHY's and Switch.
4	Power plane 1. Various power planes.
5	Power plane 2. Mainly 3V3 and DDR4L power planes.
6	Signal plane 2. Signal traces for DDR4L, RGMII to Single PHY.
7	Ground plane 2.
8 (Bottom)	LAN9696 signal traces for DDR4L. SerDes Rx paths to QuadPHy's and SFP.

The ground shield serves as a "quiet" ground for PHY copper media signals and SFP cages. It couples capacitively to the ground plane, providing a low-impedance return path for high-frequency noise.

A.2.1 Layer 1 – Top Plane

FIGURE A-2: LAYER 1 – TOP PLANE

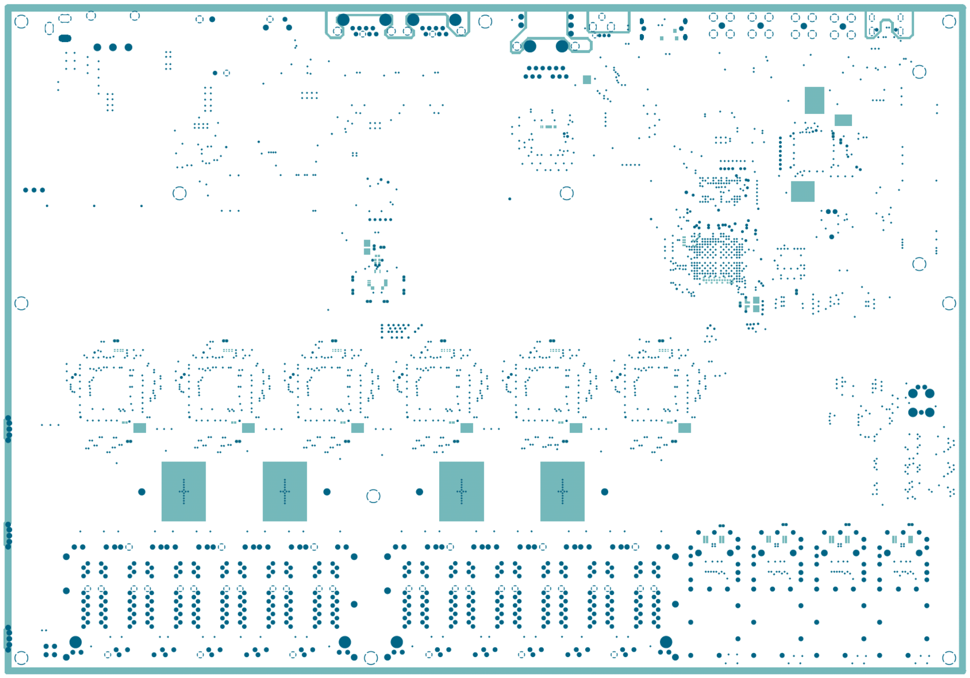


Mostly signal traces for DDR4 and SerDes Tx Macros.

12V and 48V/56V input power and PoE center tap power distribution.

A.2.2 Layer 2 – Ground Plane 1

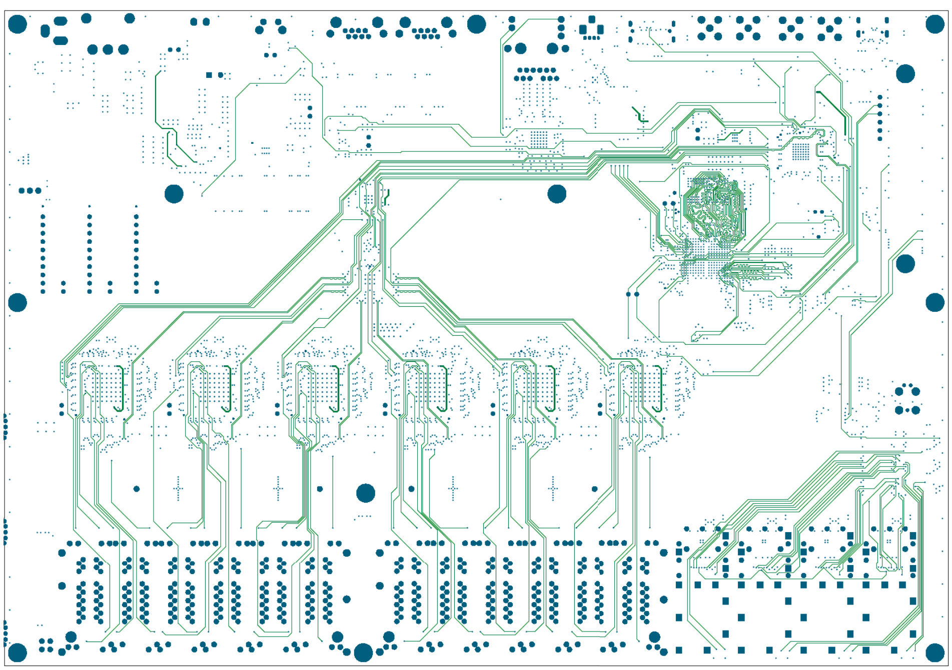
FIGURE A-3: LAYER 2 – GROUND PLANE 1



Solid ground plane.

A.2.3 Layer 3 – Signal Plane 1

FIGURE A-4: LAYER 3 – SIGNAL PLANE 1

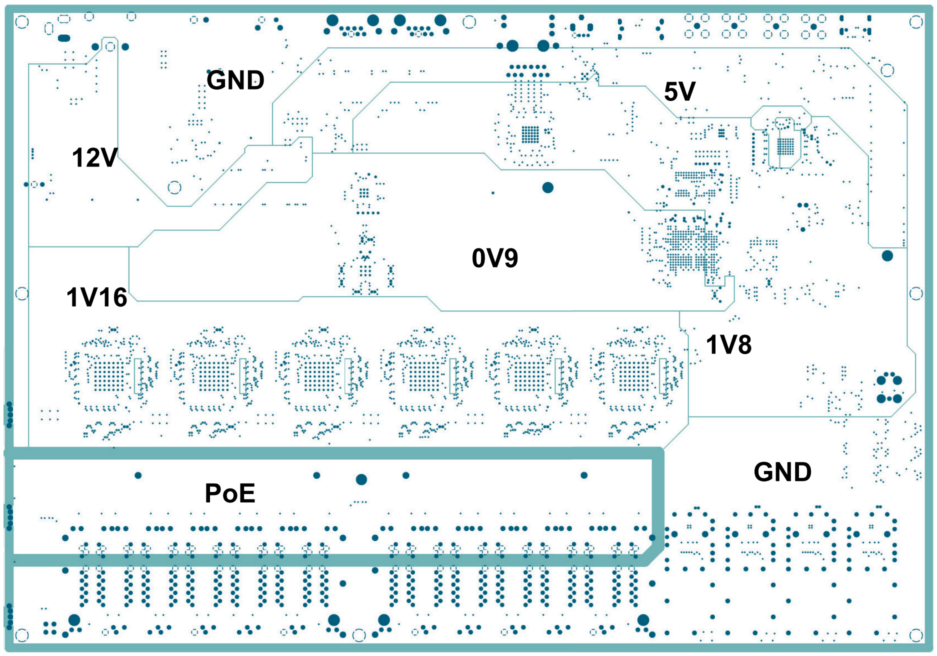


Signal traces for DDR4L. Various clocks for CuPHYs and Switch. SFP control signals.

A.2.4 Layer 4 – Power Plane 1

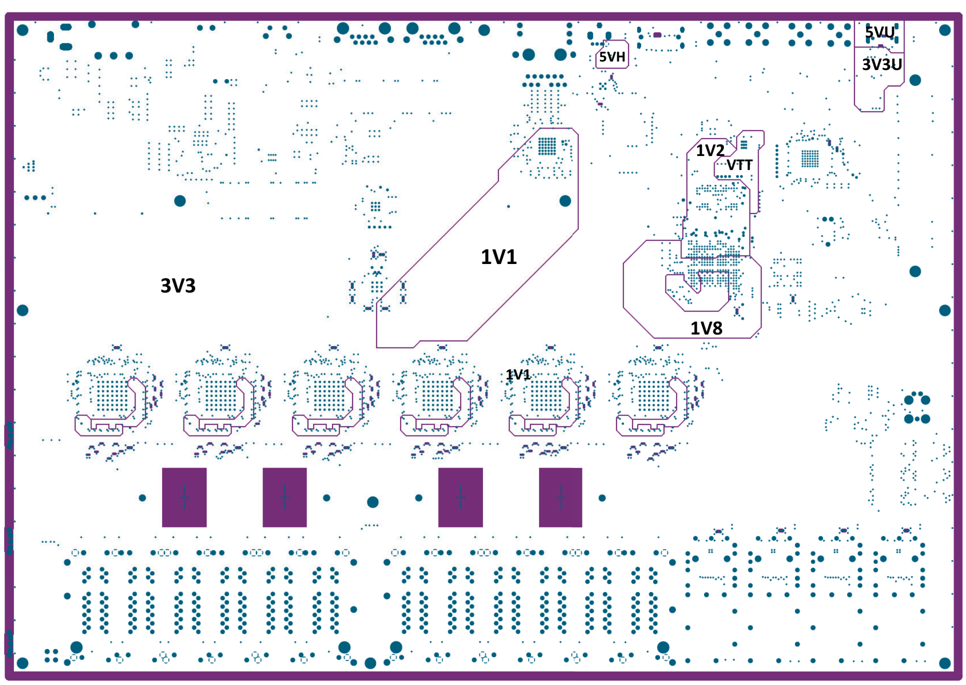
Various power planes. See Figure A-5.

FIGURE A-5: LAYER 4 – POWER PLANE 1



A.2.5 Layer 5 – Power Plane 2

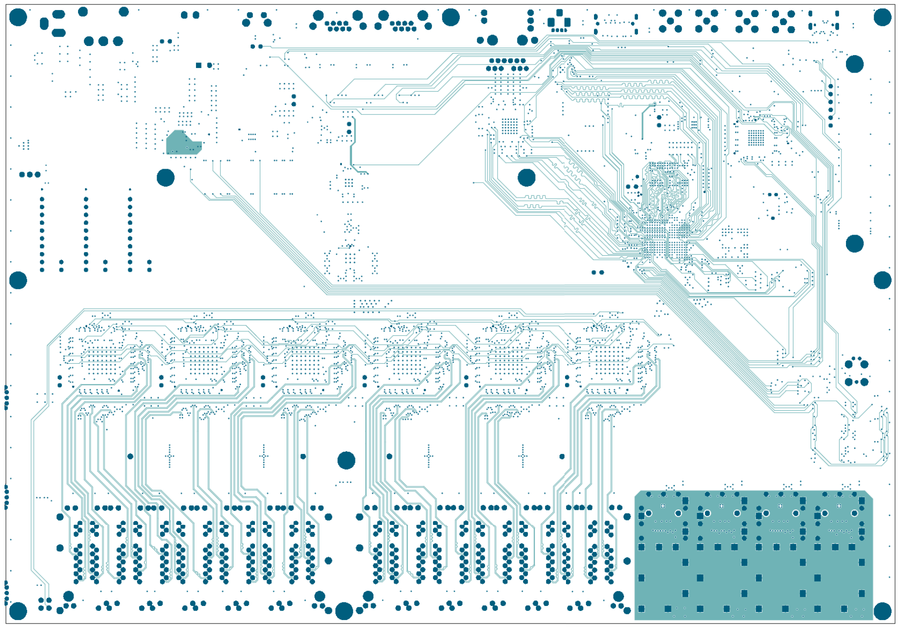
FIGURE A-6: LAYER 5 – POWER PLANE 2



Power planes; mainly the 3V3. The 1V1 is made ‘short and wide’ as possible. DDR4L 1V2 and VTT power plane. 5V Host and USB power planes.

A.2.6 Layer 6 – Signal Plane 2

FIGURE A-7: LAYER 6 – SIGNAL PLANE 2

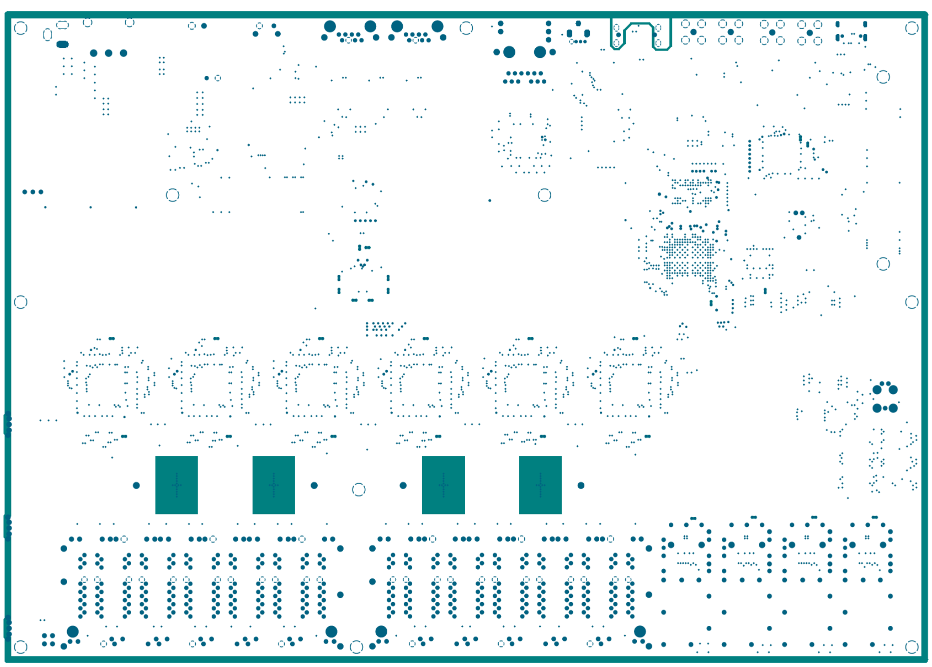


The plane is the SFP shield ground. Signal traces for DDR4, RGMII to Single PHY.

ULPI host interface.

A.2.7 Layer 7 – Ground Plane 2

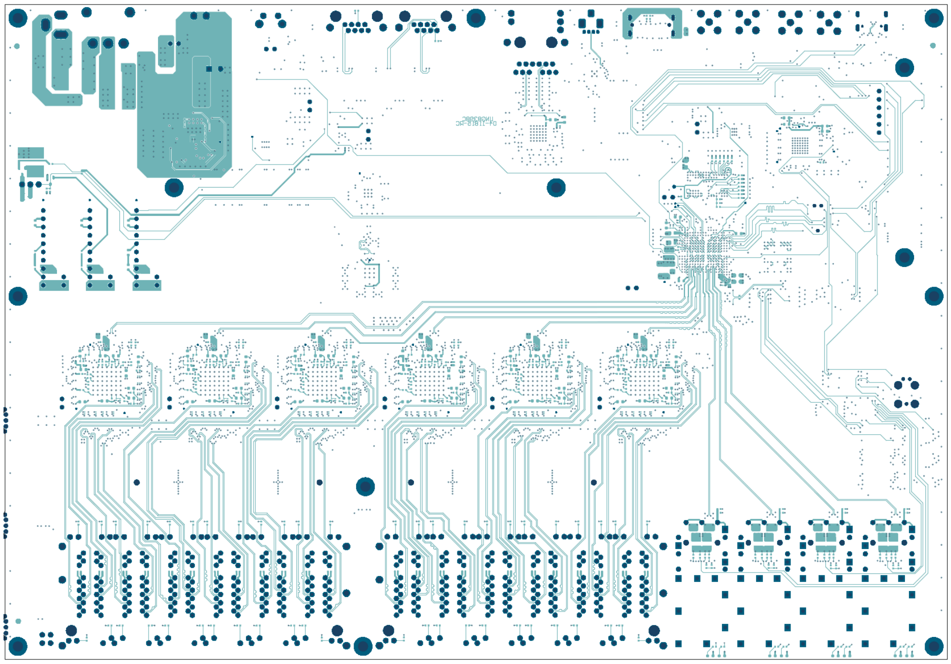
FIGURE A-8: LAYER 7 – GROUND PLANE 2



Solid ground plane.

A.2.8 Layer 8 – Bottom Plane

FIGURE A-9: LAYER 8 – BOTTOM PLANE



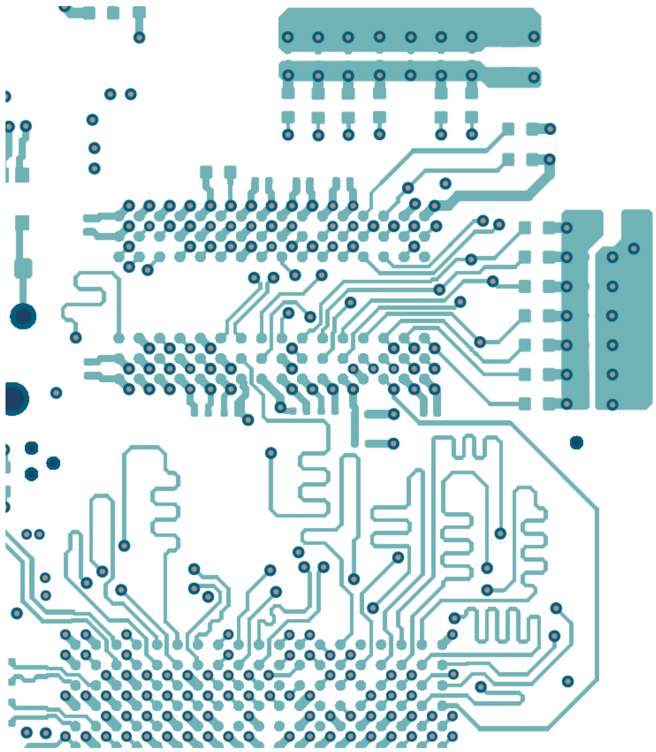
Signal traces for DDR4L and SerDes Rx macros.

Copper PHY MDI

A.2.9 DDR4 Close-up on Top Layer

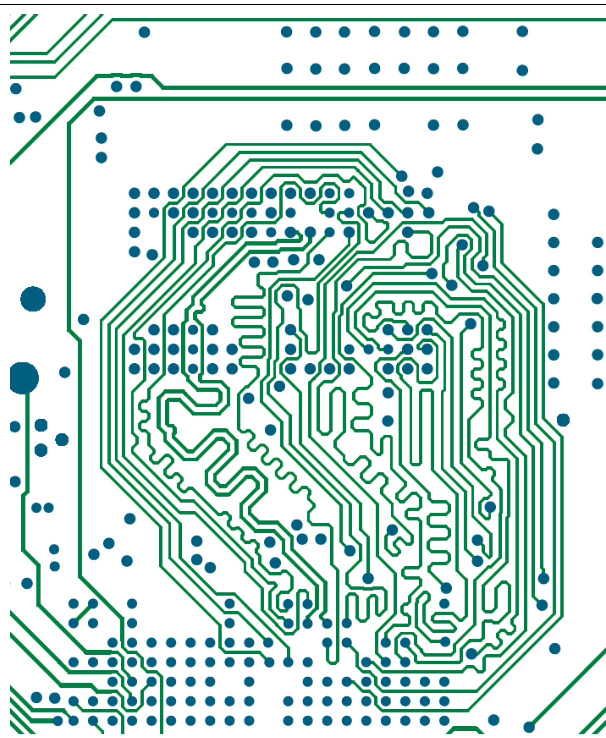
DDR4 in close-up to illustrate how the DDR length matching is acquired.

FIGURE A-10: DDR CLOSE-UP ON TOP LAYER



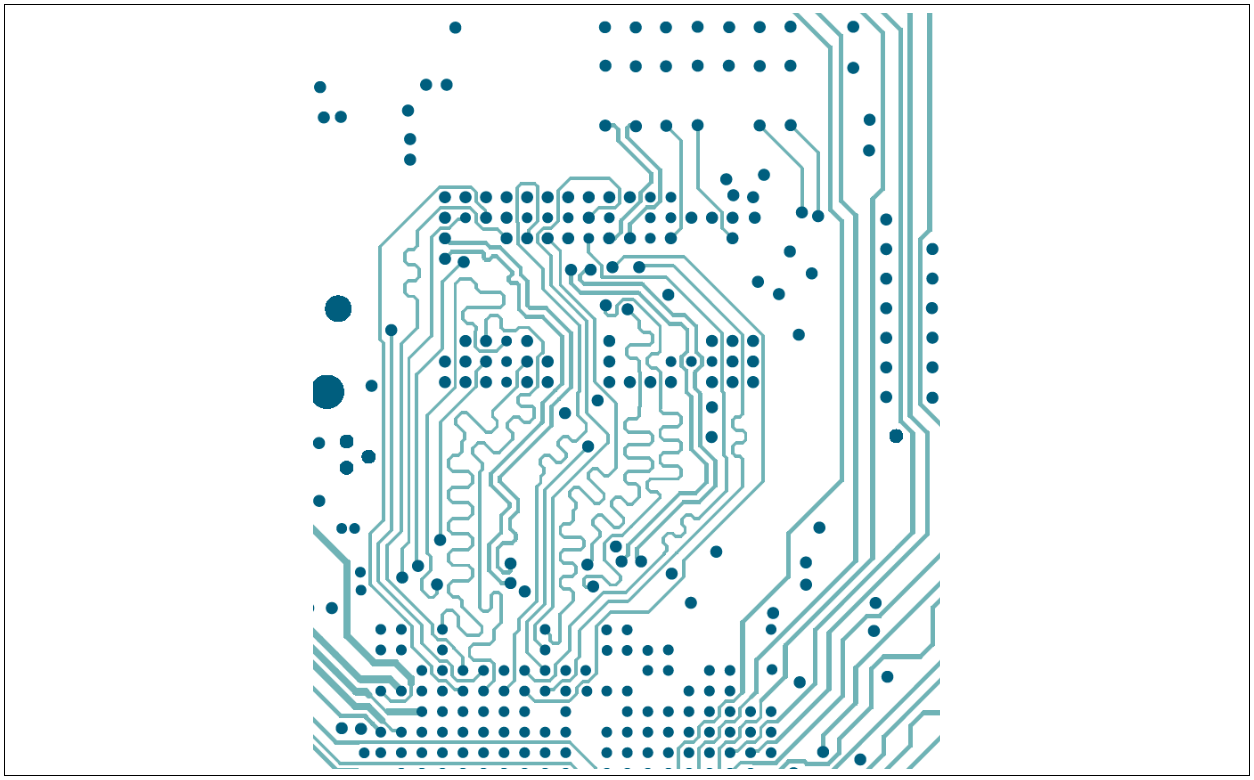
A.2.10 DDR4 Close-up on Signal Plane 1

FIGURE A-11: DDR CLOSE-UP ON SIGNAL PLANE 1



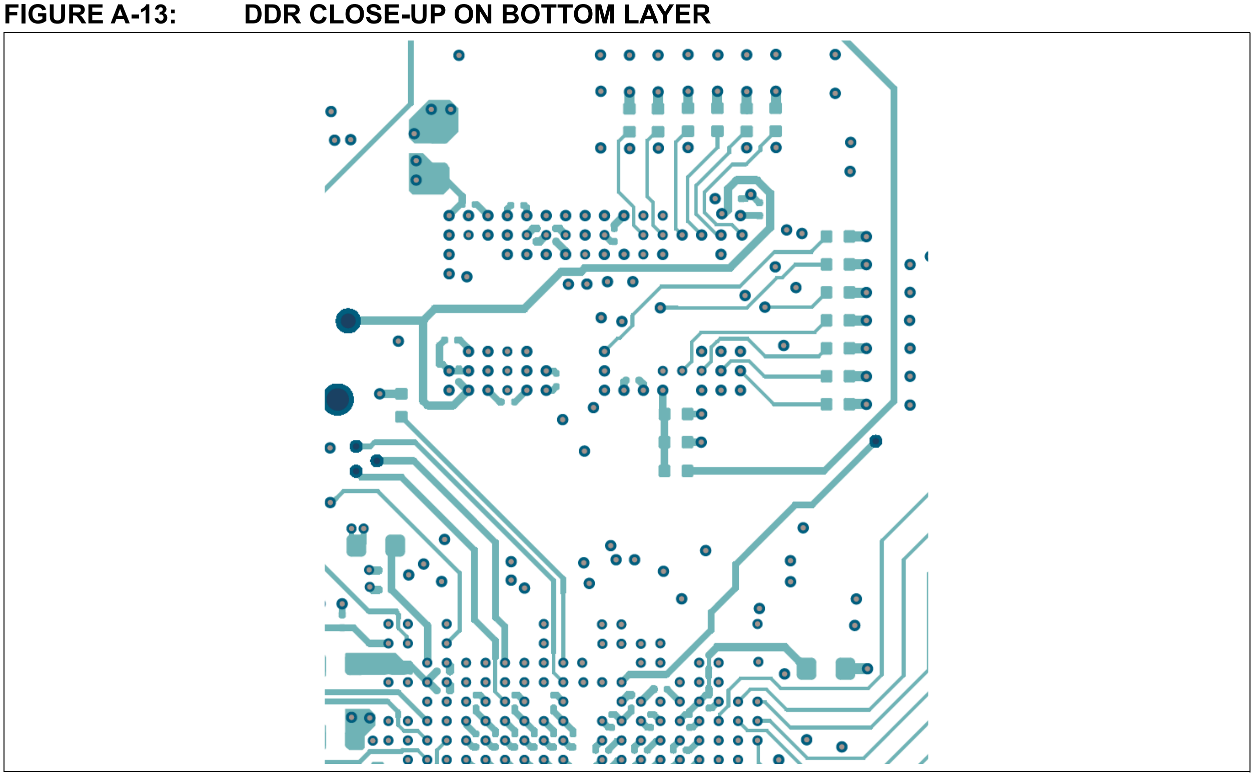
A.2.11 DDR4 Close-up on Signal Plane 2

FIGURE A-12: DDR CLOSE-UP ON SIGNAL PLANE 2



A.2.12 DDR4 Close-up on Bottom Layer

FIGURE A-13: DDR CLOSE-UP ON BOTTOM LAYER

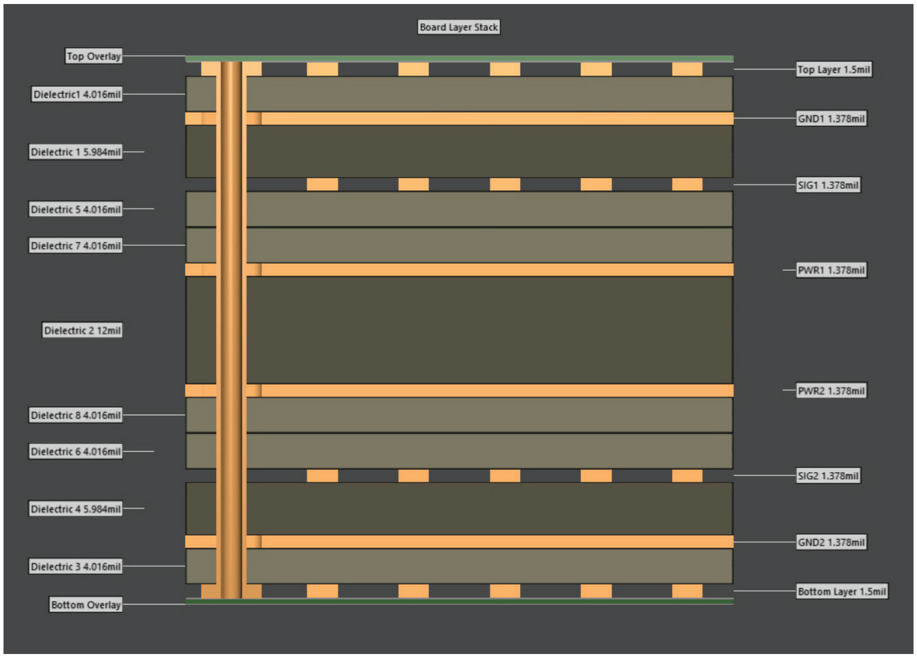


LAN9696 package skew report is available upon request. Contact your Microchip representative.

A.3 PCB LAYER STACK-UP

The reference board is an eight-layer impedance-controlled PCB. The stack-up is shown in Figure A-14.

FIGURE A-14: LAYER STACKED VISUALIZED



A.3.1 PCB Trace Widths, Clearance, and Impedance

TABLE A-2: PCB TRACE WIDTHS, CLEARANCE, AND IMPEDANCE

Layer	Name	Material	Thicknes s	Constant Dk/Df	Impedance Width/Spacing
—	Top overlay	—	—	—	—
—	Top solder	LPI Green	0.500mil	3.5, 0.025	—
—	Top finish	ENIG_3-6um Ni, 0.05-0.1um Au.	0.160mil	—	—
1	Top layer	CF-003	1.500mil	—	50Ω SE: 7.5mil 60Ω SE: 5.0mil 90Ω Diff: 6.0/4.5m 100Ω Diff: 5.0/5.0, 6.0/8.5
—	Dielectric1	PP-FR408HR-2313_56	4.016mil	3.65, 0.009	—
2	GND1	CF-004	1.378mil	—	—
—	Dielectric 1	CR-FR408HR-2x1086	5.984mil	3.6, 0.0091	—
3	SIG1	CF-004	1.378mil	—	50Ω S: 6.0mil 60Ω SE: 4.0mil 100Ω Diff: 5.0/8.5mil
-	Dielectric 5	PP-FR408HR-1080_71	4.016mil	3.34, 0.01	—
-—	Dielectric 7	PP-FR408HR-1080_71	4.016mil	3.34, 0.01	—
4	PWR1	CF-004	1.378mil	—	—
—	Dielectric 2	CR-FR408HR-28mil	12.00mil	3.7, 0.0088	—
5	PWR2	CF-004	1.378mil	—	—
—	Dielectric 8	PP-FR408HR-1080_71	4.016mil	3.34, 0.01	—
—	Dielectric 6	PP-FR408HR-1080_71	4.016mil	3.34, 0.01	—

TABLE A-2: PCB TRACE WIDTHS, CLEARANCE, AND IMPEDANCE (CONTINUED)

Layer	Name	Material	Thickness	Constant Dk/Df	Impedance Width/Spacing
6	SIG2	CF-004	1.378mil	—	50Ω SE: 6.0mil 60Ω SE: 4.0mil 100Ω Diff: 5.0/8.5mil
—	Dielectric 4	CR-FR408HR-2x1086	5.984mil	3.6, 0.0091	—
7	GND2	CF-004	1.378mil	—	—
—	Dielectric 3	PP-FR408HR-2313_56	4.016mil	3.65, 0.009	—
8	Bottom layer	CF-003	1.500mil	—	50Ω SE: 7.5mil 60Ω SE: 5.0mil 90Ω Diff: 6.0/4.5mil 100Ω Diff: 5.0/5.0, 6.0/8.5
—	Bottom finish	ENIG_3-6um Ni, 0.05-0.1um Au.	0.160mil	—	—
—	Bottom solder	LPI-Green	0.500mil	3.5, 0.025	—
—	Bottom overlay	—	—	—	—
—	Board thickness	(1.54mm)	60.652mil	—	—

TABLE A-3: NET IMPEDANCE AND LENGTH MATCHING

Net Group	Type	Impedance	Length matching	Tolerance (mm)	Max vias	Via type	Layers	Notes
Clock	SE	50Ω	None	—	4	All	All	—
SI	SE	50Ω	None	—	4	All	All	Daisy-chain
DDR_CMD_LANE	SE/DIFF	60Ω	Within lane	1	2	All	All	2
DDR_DQ_LANE0	SE/DIFF	60Ω	Within lane	1	2	All	All	2
DDR_DQ_LANE1	SE/DIFF	60Ω	Within lane	1	2	All	All	2
USB	DIFF	90Ω	P/N only	1	2	All	All	—
DiffClock	DIFF	100Ω	P/N only	1	4	All	All	—
DiffClockCrit	DIFF	100Ω	P/N only	1	2	All	All	5
SerDes10G	DIFF	100Ω	P/N only	0	2	-	Outer	1
Sense	Analog	—	—	—	—	All	All	—
Power	Power	—	—	—	—	—	—	—
Static	SE	—	—	—	—	—	—	3
Unspecified	SE	60Ω	—	—	—	—	All	4

Note 1: Differential signal on outer layers.
2: DDR SE impedance 60Ω between devices.
3: No impedance => 4 mil trace on any layer.
4: Any unspecified nets should be routed as 60Ω.
5: Critical clock nets: CLK156 - route with extra clearance.

NOTES:

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